

Al-Usra Hajj and Umrah Company Management System

Database Project Report

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1. Abstract

The Al-Usra Hajj and Umrah Company Management System is a desktop database application designed to manage the main operations of a Hajj and Umrah company. The system provides organized management for users, employees, Hajj applicants, campaigns, packages, bookings, hotels, transportation, employee tasks, and internal messages.

The project was developed using Java Swing for the graphical user interface and MySQL as the relational database management system. The application supports three main user roles: Manager, Employee, and Customer. Each role has a different dashboard and different permissions based on the tasks that the user needs to perform.

The system applies important database concepts such as primary keys, foreign keys, relationships, constraints, normalization, CRUD operations, and JDBC connectivity. It also provides a practical example of how a real company system can be designed by connecting a Java desktop application with a structured relational database.

Overall, the project aims to reduce manual work, improve data organization, support role-based access, and make the management of Hajj and Umrah services easier, faster, and more reliable.

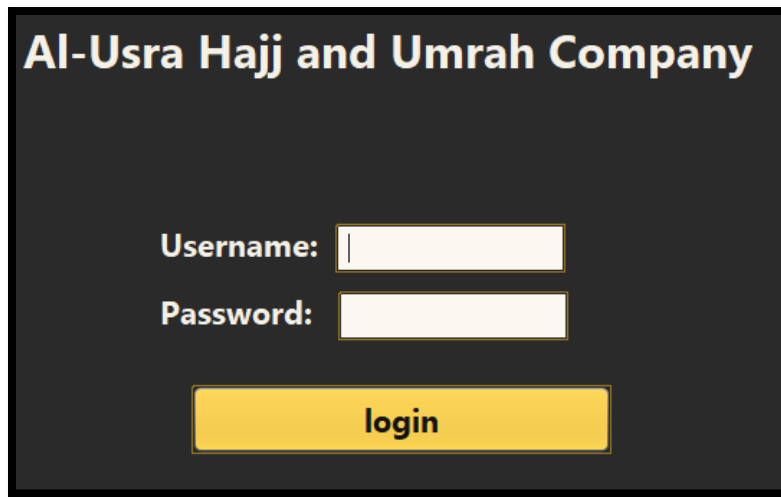
2.Introduction

Hajj and Umrah companies manage many operations that require accurate organization and continuous follow-up. These operations include registering customers, managing Hajj applications, assigning employees, preparing packages, managing bookings, organizing hotels, arranging transportation, assigning employee tasks, and sending internal messages.

Managing all these operations manually can be difficult and time-consuming. It can also lead to errors, repeated data, missing records, and delays in updating information. For this reason, a computerized database system is needed to organize the company's work and make the management process more efficient.

The Al-Usra Hajj and Umrah Company Management System was created to solve these issues by using a centralized relational database and a user-friendly desktop interface. The system allows different users to access different features based on their roles. The Manager can control the main company data, the Employee can manage tasks and messages, and the Customer can view personal Hajj-related services.

This project combines database design and Java programming. The database stores the system data in structured tables, while the Java Swing application provides graphical screens for users to interact with the data easily.



The image shows a login window for the 'Al-Usra Hajj and Umrah Company'. The window has a dark grey background with a thin black border. At the top, the company name 'Al-Usra Hajj and Umrah Company' is written in a bold, white, sans-serif font. Below the title, there are two input fields: 'Username:' followed by a white rectangular box, and 'Password:' followed by a white rectangular box. At the bottom center, there is a yellow rectangular button with the word 'login' in black, bold, sans-serif font.

Figure 1: Login interface of the Al-Usra Hajj and Umrah Company Management System.

3.Problem Statement

Hajj and Umrah companies deal with a large amount of data during their daily operations. This data includes customer information, Hajj applications, employee records, hotel reservations, transportation details, Hajj packages, bookings, employee tasks, and internal messages. If this data is managed manually, the company may face many problems such as duplicated records, missing information, slow searching, and difficulty in updating data.

Another important problem is the lack of clear access control. In a real company, not all users should have the same permissions. The Manager should be able to manage the main company records, while Employees should only access their work-related information such as tasks and messages. Customers should only be able to view their own services such as booking, hotel, and transport information.

Manual management also makes it difficult to track the status of Hajj applicants, bookings, hotel reservations, and transportation assignments. For example, if the status of a booking changes, it should be updated quickly and reflected in the customer interface. Without a database system, tracking these changes can become slow and inaccurate.

Therefore, this project was developed to solve these problems by creating a centralized database system connected to a Java desktop application. The system helps organize data, reduce errors, control user access, and make company operations easier to manage.

4.Project Objectives

The main objective of this project is to design and implement a database system that supports the management of a Hajj and Umrah company. The system aims to organize the company's data and provide an easy way to access, update, and manage information through a graphical user interface.

The project objectives are:

1. To design a relational database for a Hajj and Umrah company.

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2. To store users, employees, Hajj applicants, campaigns, bookings, hotels, transportation, tasks, and messages in organized database tables.
3. To create a Java Swing desktop application connected to the database.
4. To provide a login system based on user roles.
5. To allow the Manager to manage the main company data.
6. To allow Employees to view their profile, tasks, and messages.
7. To allow Customers to view their profile, booking, hotel, and transport information.
8. To apply CRUD operations such as adding, viewing, updating, and deleting records.
9. To use primary keys and foreign keys to maintain relationships between tables.
10. To improve the accuracy, organization, and speed of company data management.

5. Project Scope

The scope of this project includes the main operations needed by a Hajj and Umrah company. The system focuses on managing data related to users, employees, Hajj applicants, campaigns, packages, bookings, hotels, transportation, employee tasks, and messages.

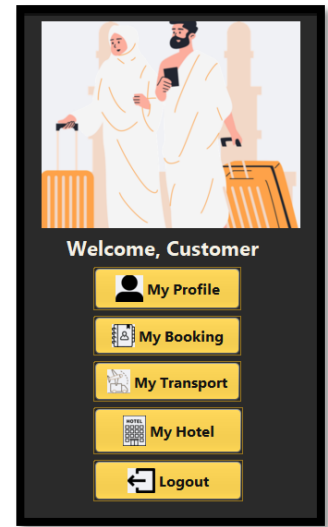
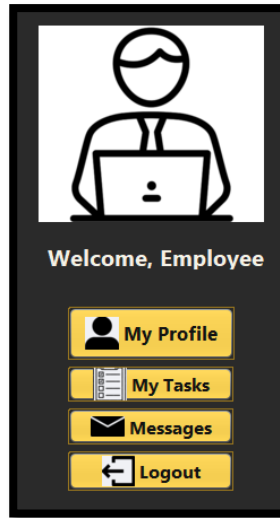
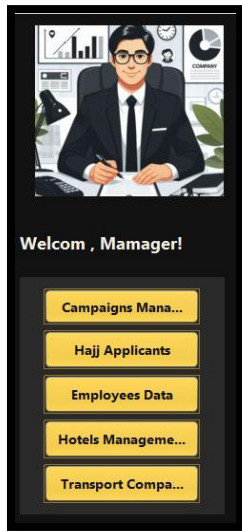
The Manager side of the system includes screens for managing campaigns, Hajj applicants, employees, hotels, and transport companies. These screens allow the Manager to add new records, update existing records, delete records, and view data in tables.

The Employee side includes screens for viewing personal profile information, checking assigned tasks, marking tasks as done, and reading or replying to messages. The Employee does not have access to the full company management screens.

The Customer side includes screens for viewing personal Hajj information, booking details, hotel reservation, and transport information. Customers have limited access and can only view or update specific personal information such as phone number and password.

The project does not cover online payment, mobile application support, or real-time notifications. These features can be added in future improvements. The current scope focuses on database organization, role-based access, CRUD operations, and Java-MySQL integration.

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Manager → Employee → Customer

6.Tools and Technologies Used

The project was developed using two main tools: **NetBeans** and **MySQL Workbench**. Each tool was used for a specific purpose during the development process.

NetBeans was used as the main development environment for building the Java desktop application. It helped in writing and organizing the Java code, designing the graphical user interfaces, and managing the project files. Through NetBeans, the system interfaces were created using Java Swing components such as buttons, labels, text fields, combo boxes, panels, and tables.

MySQL Workbench was used to design, create, and manage the database. It helped in creating tables, defining primary keys and foreign keys, inserting sample data, testing SQL queries, and viewing table contents. It was also used to export the database as a SQL file for submission.

The project also used **Java Swing** to build the graphical user interface, **MySQL** as the relational database management system, and **JDBC** to connect the Java application with the MySQL database. Together, these tools allowed the system to store data, retrieve records, update information, and display results inside the application interfaces.

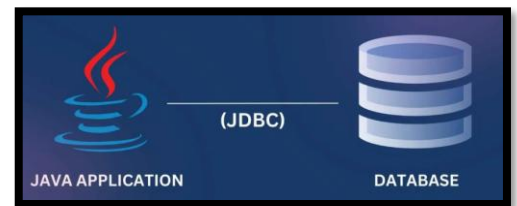


Figure: Tools and technologies used in the project.

7. System Users and Roles

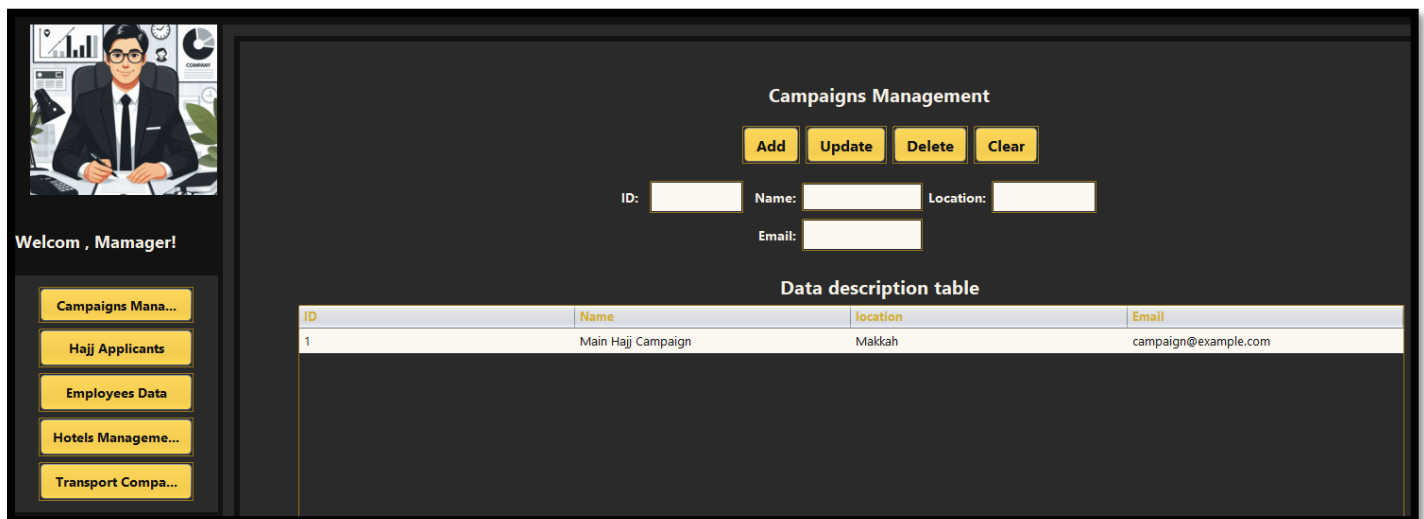
The system is designed based on three main user roles: **Manager**, **Employee**, and **Customer**. Each role has different permissions and different interfaces. This role-based design helps organize the system and prevents users from accessing features that are not related to their responsibilities.

The **Manager** has the highest level of access in the system. The Manager can manage the main company data, including campaigns, Hajj applicants, employees, hotels, and transport companies. The Manager can add, update, delete, and view records from the management screens.

The **Employee** has limited access compared to the Manager. The Employee can view personal profile information, check assigned tasks, mark tasks as done, and read or reply to internal messages. The Employee dashboard focuses on work-related features only.

The **Customer** has access to personal Hajj-related services. The Customer can view profile information, booking details, hotel reservation, and transport information. The Customer can only edit limited information such as phone number and password.

This separation of roles makes the system more secure, organized, and easier to use.



The screenshot displays the Manager's dashboard. On the left is a sidebar with a user profile and a list of management options. The main area is titled 'Campaigns Management' and includes form fields for adding or updating campaign records, along with a table showing existing data.

Campaigns Management

Buttons: Add, Update, Delete, Clear

Form fields: ID, Name, Location, Email

Data description table

ID	Name	location	Email
1	Main Hajj Campaign	Makkah	campaign@example.com

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Welcome, Employee

- My Profile
- My Tasks
- Messages
- Logout



View & Update Your Information

My Profile

First Name: Second Name: Email:

[Update Profile](#) [Reset](#) [Back to Dashboard](#) [Sort by Salary](#)

[Logout](#)

My Information

Employee_id	first_name	last_name	Email	Gender	Salary
1	bb	noor	ahmad@example.com	Male	700.0



Welcome, Customer

- My Profile
- My Booking
- My Transport
- My Hotel
- Logout



You can only edit Phone and Password

My Profile

[Edit Profile](#) [Change Password](#)

[Logout](#)

Hajji Information	Supervisor Information	Country Information
1 bb User Male 2000-01-01 45454545 Pending	osama supervisor@example.com 599000000 5	Palestine ILS/JOD Arabic Asia

Manager | Employee | Customer

8. System Features

The system provides several important features that support the daily operations of the Hajj and Umrah company. These features were designed to make data management easier and more organized for different users.

One of the main features is the **role-based login system**. The system checks the username and password, then opens the correct dashboard based on the user role. This allows each user to access only the features related to their role.

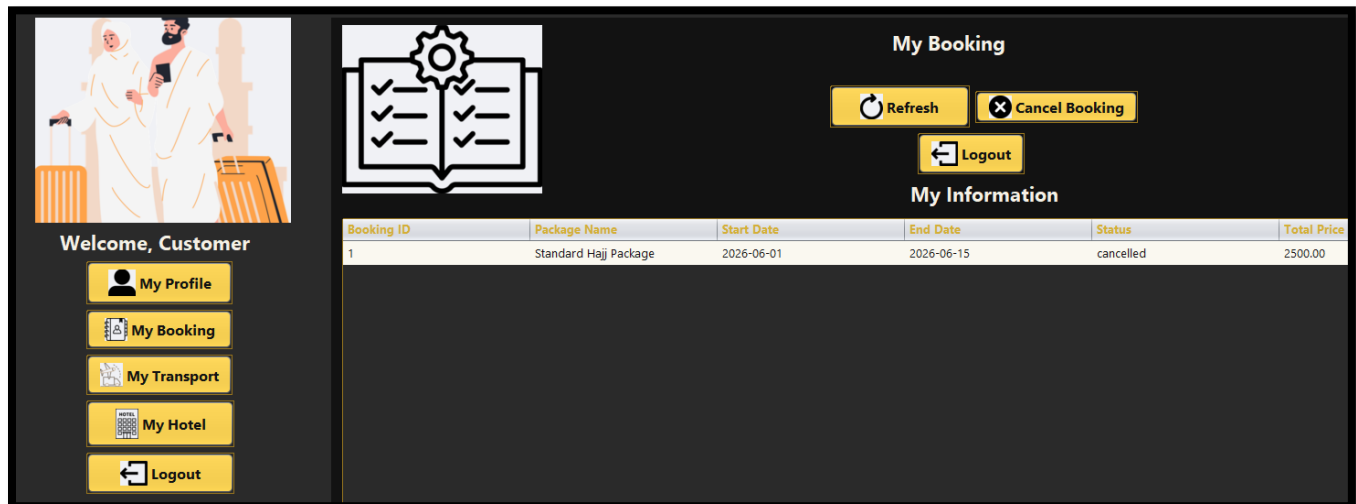
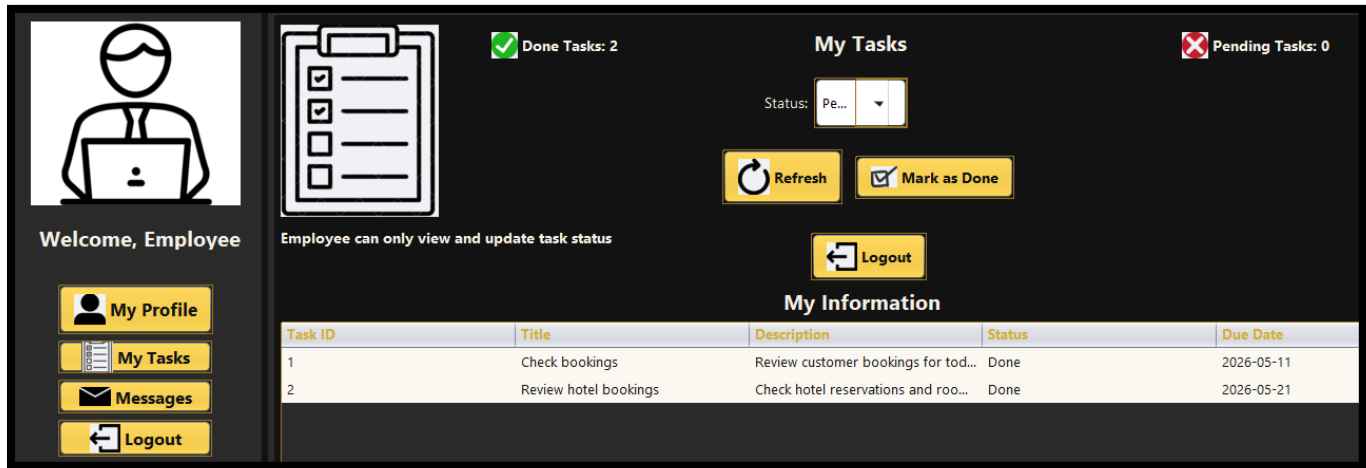
The system also supports **CRUD operations**, which include creating, reading, updating, and deleting records. These operations are used in the Manager dashboard to manage campaigns, Hajj applicants, employees, hotels, and transport companies.

Another important feature is **employee task management**. Employees can view their assigned tasks, check task status, and mark tasks as done. The system also shows the number of pending and completed tasks.

The system includes **internal messaging** between the Manager and Employees. This helps improve communication inside the company. Employees can read messages, mark them as read, and send replies.

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For Customers, the system provides access to Hajj-related services such as profile information, booking details, hotel reservation, and transport information. This allows customers to follow their services in an organized way.



9. Features Summary Table

Feature	Description	Main User
Role-Based Login	Opens the correct dashboard according to the user role.	All Users
Campaign Management	Add, update, delete, and view Hajj campaigns.	Manager
Hajj Applicants Management	Manage pilgrims and application status.	Manager
Employee Management	Manage employee records and send messages.	Manager
Hotel Management	Manage hotel records used by the company.	Manager

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Transport Company Management	Manage transportation company records.	Manager
Task Management	View tasks and mark them as done.	Employee
Internal Messages	Read, reply, and mark messages as read.	Employee
Booking View	View package booking information.	Customer
Hotel and Transport View	View hotel and transport service details.	Customer

10. Functional Requirements

Functional requirements describe the main functions that the system must provide to its users. In this project, the functions are divided according to the user role: Manager, Employee, and Customer.

The system must allow users to log in using a valid username and password. After login, the system must identify the user role and open the correct dashboard. The Manager should be able to manage the main company data, including campaigns, Hajj applicants, employees, hotels, and transport companies.

The system must allow the Manager to add new records, update existing records, delete records, and view data in tables. These operations are important because they allow the company to keep its information updated and organized.

The system must allow Employees to view their profile information, view assigned tasks, mark tasks as done, and read or reply to messages. Employees should only see information related to their work and should not have full access to management screens.

The system must allow Customers to view their Hajj profile, booking details, hotel reservation, and transport information. Customers should have limited access and should only be able to update specific information such as phone number and password.

11. Functional Requirements Table

User Role	Functional Requirement
All Users	The system shall allow users to log in using username and password.
All Users	The system shall open the correct dashboard based on user role.
Manager	The system shall allow the Manager to manage campaigns.

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Manager	The system shall allow the Manager to manage Hajj applicants.
Manager	The system shall allow the Manager to manage employee records.
Manager	The system shall allow the Manager to manage hotels.
Manager	The system shall allow the Manager to manage transport companies.
Manager	The system shall allow the Manager to send messages to employees.
Employee	The system shall allow Employees to view and update profile information.
Employee	The system shall allow Employees to view assigned tasks.
Employee	The system shall allow Employees to mark tasks as done.
Employee	The system shall allow Employees to read and reply to messages.
Customer	The system shall allow Customers to view personal Hajj information.
Customer	The system shall allow Customers to view booking information.
Customer	The system shall allow Customers to view hotel information.
Customer	The system shall allow Customers to view transport information.

12. Non-Functional Requirements

Non-functional requirements describe the quality characteristics of the system. These requirements focus on how the system should perform rather than what the system should do.

The system should be easy to use. The graphical interfaces should be clear and organized so that users can understand the available actions quickly. Buttons, tables, and labels should be placed in a way that makes the system simple to navigate.

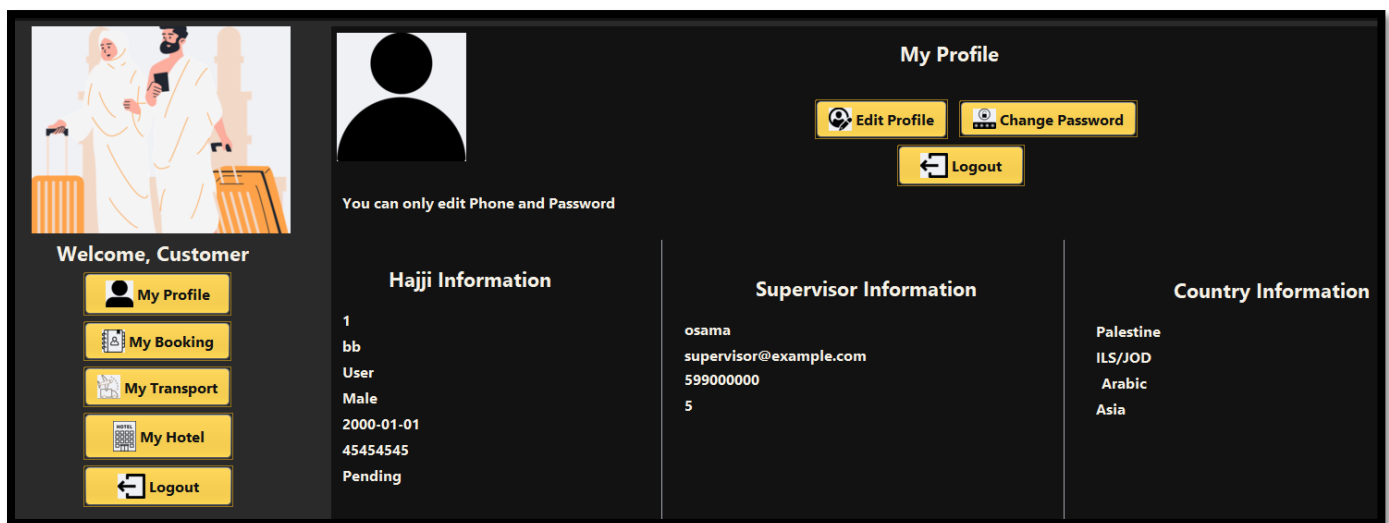
The system should be reliable. Data should be stored correctly in the database, and the interfaces should display updated information after operations such as adding, updating, deleting, or refreshing records.

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The system should maintain data integrity. This is achieved through primary keys, foreign keys, constraints, and validation rules. These database rules help prevent invalid data and maintain correct relationships between tables.

The system should also provide basic security through role-based access. Each user should only access the functions related to their role. The Manager, Employee, and Customer dashboards are separated to prevent unauthorized access to unrelated features.

The system should be maintainable. The project is divided into different classes and panels, which makes the code easier to understand, update, and improve in the future.



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Figure: Organized dashboard design for better usability.

13. Non-Functional Requirements Table

Requirement	Description
Usability	The system should provide clear and easy-to-use interfaces.
Reliability	The system should store and display data correctly.
Data Integrity	The database should prevent invalid or inconsistent records.
Security	Users should access only the features related to their role.
Maintainability	The code should be organized into classes and panels.
Performance	The system should load data from the database in a reasonable time.
Availability	The application should be available on the local computer when the database is running.

14. Business Rules

Business rules describe the conditions and restrictions that must be followed by the system. These rules help ensure that the system behaves in a logical way according to the needs of a Hajj and Umrah company.

Each user must have one role, such as Manager, Employee, or Customer. The role determines which dashboard the user can access. A Manager can manage company records, while an Employee and a Customer have limited access.

Each Hajj applicant represents a customer who applied for Hajj services. A Hajj applicant can be connected to bookings, hotel reservations, transport assignments, country information, and supervisor information. This makes the hajj_applicants table one of the central tables in the database.

Each employee can have many assigned tasks. A task belongs to one employee and contains information such as title, description, status, and due date. The employee can mark the task as done when it is completed.

Each booking must be connected to a valid Hajj applicant and a valid package. Hotel bookings must be connected to a valid Hajj applicant and a valid hotel. Transport assignments must be connected to a valid Hajj applicant and a valid transport record.

These rules help the system keep data organized and make sure that records are connected correctly.



booking_id	hajj_id	package_id	status	total_price
1	1	1	cancelled	2500.00
NULL	NULL	NULL	NULL	NULL

Figure: Sample data showing business rules between applicants and bookings.

15. Business Rules Summary Table

Rule	Description
User Role Rule	Each user must have one role: Manager, Employee, or Customer.
Applicant Rule	Each Hajj applicant represents one customer in the system.
Booking Rule	Each booking must be linked to a valid Hajj applicant and package.
Hotel Booking Rule	Each hotel booking must be linked to a valid applicant and hotel.
Transport Rule	Each transport assignment must link an applicant with a transport record.
Employee Task Rule	Each task belongs to one employee.
Message Rule	Messages are linked to employees and used for internal communication.
Status Rule	Bookings, tasks, and applications use status values to track progress.

16. Database Overview

The database is the main part of the system because it stores all information used by the application. The project uses a relational database in MySQL, where data is divided into tables. Each table stores a specific type of information, and tables are connected using primary keys and foreign keys.

The database contains tables for users, employees, Hajj applicants, campaigns, packages, bookings, hotels, hotel bookings, transport companies, transports, employee tasks, and employee messages. This separation makes the database easier to manage and reduces duplication.

For example, employee information is stored in the employees table, while employee tasks are stored in the employee_tasks table. The two tables are connected using emp_id. This means that each task belongs to a specific employee without repeating the employee information inside the task table.

The database design supports the main functions of the system. The Manager uses tables related to company management, Employees use tables related to tasks and messages, and Customers use tables related to bookings, hotels, and transport services.

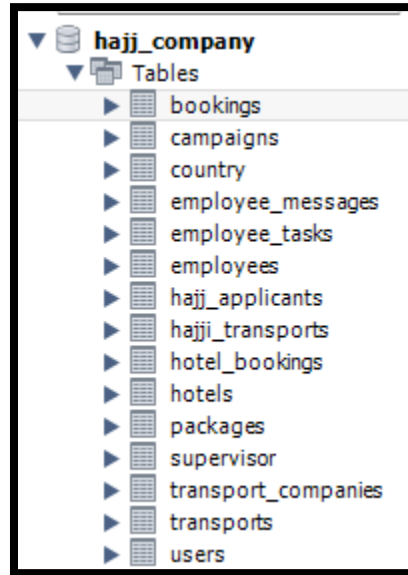


Figure: Database overview showing the main tables in MySQL Workbench.

17. Main Database Tables

The database consists of several tables, and each table has a specific purpose. The following tables are the most important tables in the system:

- users: stores login accounts and user roles.
- employees: stores employee information.
- hajj_applicants: stores customer and pilgrim information.
- campaigns: stores Hajj campaign information.
- packages: stores Hajj package details.
- bookings: stores package booking records.
- hotels: stores hotel information.
- hotel_bookings: stores hotel reservation records.
- transport_companies: stores transport company information.
- transports: stores transport trip details.
- hajji_transports: connects Hajj applicants with transport trips.
- employee_tasks: stores employee tasks.

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- employee_messages: stores internal messages between management and employees.

These tables are connected together to support the complete system workflow. For example, a customer is stored as a Hajj applicant, then this applicant can be connected to a booking, a hotel reservation, and a transport assignment.

	id	username	password	role
▶	1	admin	admin123	admin
	2	employee	emp123	employee
	3	customer	cust123	customer
✱	NULL	NULL	NULL	NULL

	hajj_id	user_id	first_name	last_name	gender	phone	dob	status	supervisor_id	country_id
▶	1	3	mohammed	jaber	Male	45454545	2000-01-01	Pending	1	1
	2	3	ali	ameen	Male	056243	2000-01-08	Pending	1	1
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

18. Main Tables Summary

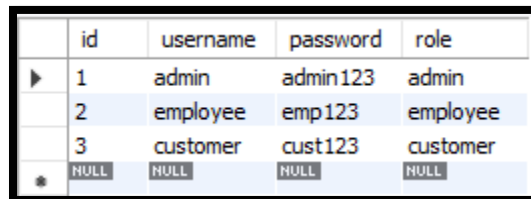
Table Name	Main Purpose
Users	Stores usernames, passwords, and roles.
Employees	Stores employee personal and work information.
hajj_applicants	Stores pilgrims or customers applying for Hajj services.
campaigns	Stores campaign information managed by the Manager.
packages	Stores Hajj package details such as name, dates, and price.
bookings	Stores customer bookings connected to packages.
hotels	Stores hotel details such as name, phone, city, and rating.
hotel_bookings	Stores hotel reservation information for Hajj applicants.
transport_companies	Stores transportation company details.
transports	Stores transport trip information.
hajji_transports	Links Hajj applicants with transport trips.
employee_tasks	Stores tasks assigned to employees.
employee_messages	Stores messages between management and employees.

19. Users Table Explanation

The users table is responsible for storing login information. It contains the username, password, and role of each user. The role determines which dashboard will open after login.

The system supports three roles: Manager, Employee, and Customer. When a user enters a username and password, the system searches this table. If the information is correct, the system checks the role and opens the correct interface.

This table is important because it controls the first step of the system, which is authentication. It helps separate users and prevents each user from accessing dashboards that are not related to their role.



	id	username	password	role
▶	1	admin	admin123	admin
	2	employee	emp123	employee
	3	customer	cust123	customer
*	NULL	NULL	NULL	NULL

Figure: Users table used for login and role management.

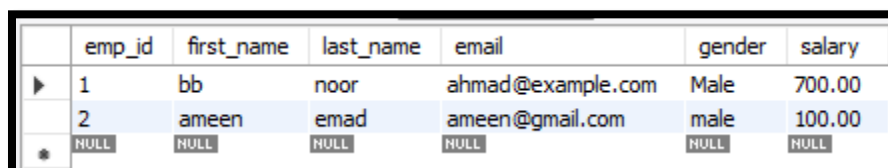
20. Employees Table Explanation

The employees table stores employee information used in the Manager and Employee dashboards. It contains data such as employee ID, first name, last name, email, gender, and salary.

The Manager can manage employee records through the Employees Data Management interface. This includes adding new employees, updating employee information, deleting employee records, and viewing employee data inside a table.

The Employee Dashboard also uses this table to display the logged-in employee profile. This allows the employee to view personal information and update limited fields depending on the system design.

The employees table is also connected to other tables such as employee_tasks and employee_messages. This means each employee can have assigned tasks and internal messages.



	emp_id	first_name	last_name	email	gender	salary
▶	1	bb	noor	ahmad@example.com	Male	700.00
	2	ameen	emad	ameen@gmail.com	male	100.00
*	NULL	NULL	NULL	NULL	NULL	NULL

21. Hajj Applicants Table Explanation

The hajj_applicants table is one of the most important tables in the database because it represents the customers or pilgrims who apply for Hajj services. It stores personal information such as Hajj applicant ID, first name, last name, gender, phone number, date of birth, and application status.

The status field is used to track the progress of each applicant. Possible values may include Pending, Approved, Rejected, Cancelled, or Completed. This helps the company follow each applicant and know the current stage of the application.

This table is connected to several other tables. A Hajj applicant can be connected to bookings, hotel bookings, transport assignments, country information, and supervisor information. Because of these connections, the table acts as a central table in the Customer side of the system.

	hajj_id	user_id	first_name	last_name	gender	phone	dob	status	supervisor_id	country_id
▶	1	3	mohammed	jaber	Male	45454545	2000-01-01	Pending	1	1
	2	3	ali	ameen	Male	056243	2000-01-08	Pending	1	1
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure: Hajj applicants table storing customer and pilgrim information.

22. Campaigns Table Explanation

The campaigns table stores information about Hajj campaigns managed by the company. Each campaign has an ID, name, location, and email. This table is managed by the Manager through the Campaigns Management interface.

The purpose of this table is to organize campaign records in one place. The Manager can add new campaigns, update campaign details, delete campaigns, and view all campaigns in a table.

The campaign ID is usually generated automatically by the database using AUTO_INCREMENT. This ensures that each campaign has a unique identifier and prevents duplicate IDs.

	id	name	location	email
▶	1	Main Hajj Campaign	Makkah	campaign@example.com
*	NULL	NULL	NULL	NULL

23. Packages and Bookings Tables Explanation

The packages table stores Hajj package information. Each package includes a package ID, package name, start date, end date, and price. Packages represent the available Hajj offers provided by the company.

The bookings table stores customer reservations for these packages. A booking connects a Hajj applicant with a specific package. It also stores the booking status and total price.

The difference between a package and a booking is important. A package is the offer itself, while a booking is a customer reservation for that offer. For example, the company may create a package called “Standard Hajj Package,” but it will appear in the Customer Booking interface only if the customer has a booking connected to that package.

This relationship helps separate package details from customer reservations and reduces repeated data.

	booking_id	hajji_id	package_id	status	total_price
▶	1	1	1	cancelled	2500.00
*	NULL	NULL	NULL	NULL	NULL

	package_id	package_name	start_date	end_date	price
▶	1	Standard Hajj Package	2026-06-01	2026-06-15	2500.00
	2	2nd package	2025-04-23	2025-06-01	24.86
*	NULL	NULL	NULL	NULL	NULL

24. Hotels and Hotel Bookings Tables Explanation

The hotels table stores hotel information used by the company. It includes details such as hotel ID, hotel name, phone number, street address, ZIP code, city, state, and rating. The Manager can manage this table through the Hotel Management interface.

The hotel_bookings table stores hotel reservation information for Hajj applicants. It connects a Hajj applicant with a specific hotel and stores room type, check-in date, check-out date, booking status, and price.

This design separates hotel information from hotel reservation information. The hotel table stores general hotel data, while the hotel bookings table stores customer-specific reservation details.

For example, one hotel can be used by many Hajj applicants. Instead of repeating the hotel name and location for every customer, the system stores hotel information once in the hotels table and connects it to reservations through hotel_id.

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	hotel_id	hotel_name	phone	street_address	zip	city	state	rating
▶	1	Makkah Hotel	123456789	Central Area	24231	Makkah	Makkah Region	5
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

	booking_id	hajji_id	hotel_id	room_type	check_in	check_out	status	price
▶	1	1	1	Double	2026-06-01	2026-06-10	cancelled	800.00
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

25. Transport Tables Explanation

The transport part of the database is divided into three main tables: transport_companies, transports, and hajji_transports.

The transport_companies table stores information about transportation companies, such as company ID, company name, email, phone number, and rating. This table is managed by the Manager through the Transport Company Management interface.

The transports table stores transport trip details. It includes information such as transport ID, company ID, vehicle type, starting location, destination, and departure date. This table represents the available transport trips.

The hajji_transports table connects Hajj applicants with transport trips. This table is important because adding a transport trip alone does not mean that it belongs to a customer. The transport trip must be linked to a Hajj applicant through hajji_transports to appear in the Customer Transport interface.

This structure makes transportation management more organized and allows one transport trip to be assigned to multiple Hajj applicants.

	transport_id	company_id	vehide_type	from_location	to_location	departure_date
▶	1	1	Bus	Jeddah Airport	Makkah	2026-06-01
	2	1	plane	palestine	makkah	2026-06-05
*	NULL	NULL	NULL	NULL	NULL	NULL

26. Employee Tasks Table Explanation

The employee_tasks table stores tasks assigned to employees. Each task has a task ID, employee ID, title, description, status, and due date.

This table is connected to the employees table using emp_id. This means that each task belongs to a specific employee. The Employee Dashboard displays only the tasks assigned to the logged-in employee.

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The task status helps employees and management track work progress. A task can be Pending or Done. When the employee finishes a task, the system updates the task status to Done.

This feature helps the company organize employee work and follow task completion.

	task_id	emp_id	title	description	status	due_date
▶	1	1	Check bookings	Review customer bookings for today	Done	2026-05-11
	2	1	Review hotel bookings	Check hotel reservations and room types	Done	2026-05-21
*	NULL	NULL	NULL	NULL	NULL	NULL

27. Employee Messages Table Explanation

The employee_messages table stores internal messages between management and employees. Each message has a message ID, employee ID, sender, message text, sent date, and status.

This table supports communication inside the company. The Manager can send messages to employees, and employees can view these messages from the Employee Dashboard. Employees can also mark messages as read and reply to messages.

The message status helps track whether a message is Read or Unread. This is useful because employees can quickly know if they have new messages.

This feature improves communication and makes the system more interactive.

	message_id	emp_id	sender	message	sent_at	status
▶	1	1	Admin	Please check the new booking requests.	2026-05-08 17:22:07	Read
	20	1	admin	hi ali	2026-05-14 10:35:45	Read
	21	1	Employee	hi admin	2026-05-14 10:41:06	Unread
*	NULL	NULL	NULL	NULL	NULL	NULL

28. Relationships Between Tables

The database is designed as a relational database, which means that the tables are connected together using relationships. These relationships are created using primary keys and foreign keys. A primary key uniquely identifies each record in a table, while a foreign key connects a record in one table to a record in another table.

Relationships are important because they help organize data and reduce duplication. Instead of repeating the same information in many tables, the system stores the information once and connects it to other tables through IDs.

For example, the employees table stores employee information, while the employee_tasks table stores tasks. The two tables are connected using emp_id. This means that each task belongs to a specific employee.

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Another example is the relationship between hajj_applicants, bookings, and packages. The bookings table connects a Hajj applicant to a package. This allows the system to know which customer booked which Hajj package.

The same idea is applied to hotel reservations and transport assignments. The hotel_bookings table connects Hajj applicants with hotels, and the hajji_transports table connects Hajj applicants with transport trips.

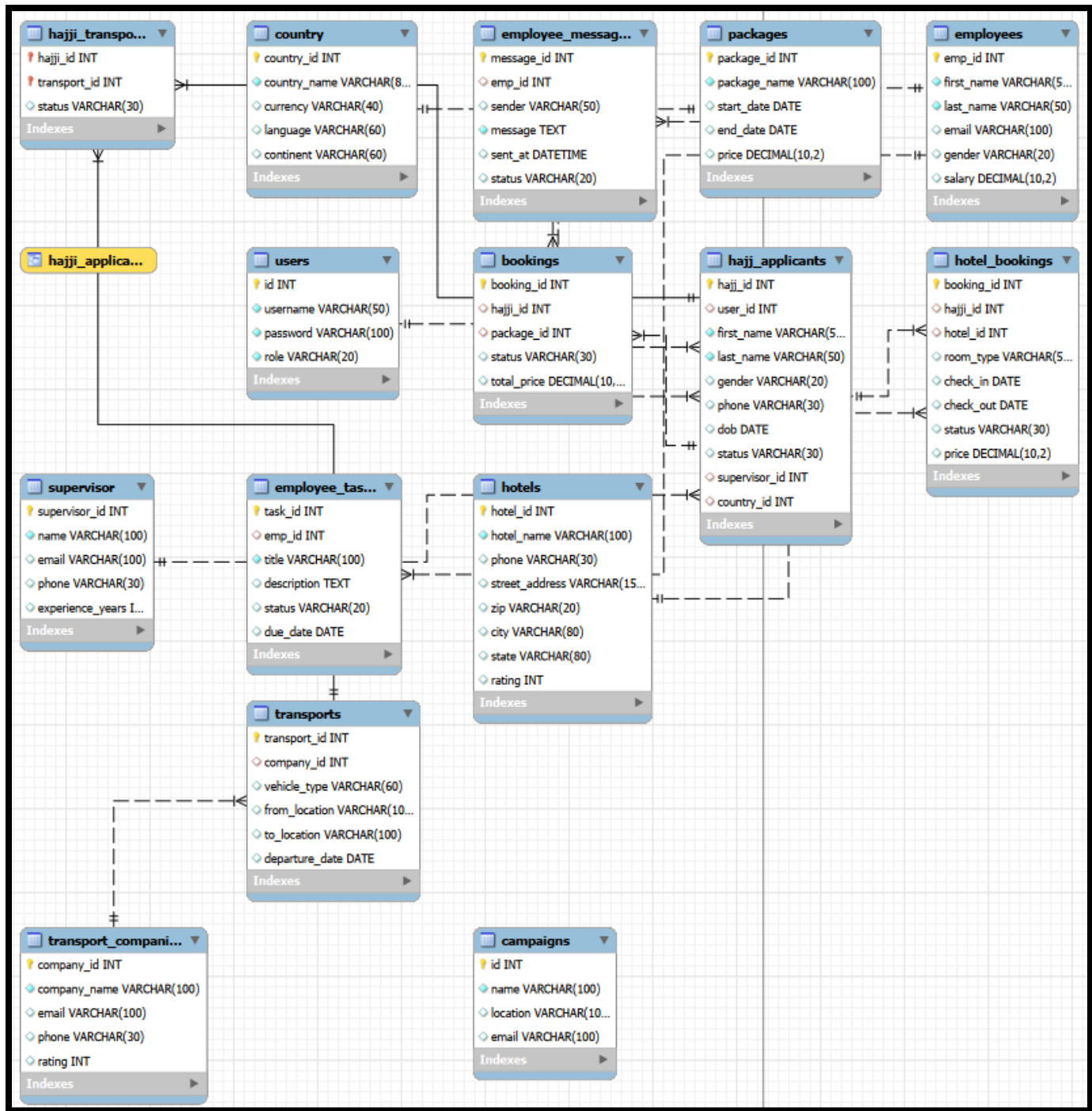


Figure: Main relationships between database tables.

29. Main Relationships Summary

The following table summarizes the main relationships used in the database:

Relationship	Description
users → hajj_applicants	A user account can be connected to a Hajj applicant account.
employees → employee_tasks	One employee can have many assigned tasks.
employees → employee_messages	One employee can have many internal messages.
hajj_applicants → bookings	One Hajj applicant can have booking records.
packages → bookings	One package can be used in many bookings.
hotels → hotel_bookings	One hotel can be used in many hotel reservations.
hajj_applicants → hotel_bookings	One Hajj applicant can have hotel booking records.
transport_companies → transports	One transport company can provide many transport trips.
transports → hajji_transports	One transport trip can be assigned to Hajj applicants.
hajj_applicants → hajji_transports	One Hajj applicant can be assigned to transport trips.
country → hajj_applicants	A country can be related to many Hajj applicants.
supervisor → hajj_applicants	A supervisor can be assigned to many Hajj applicants.

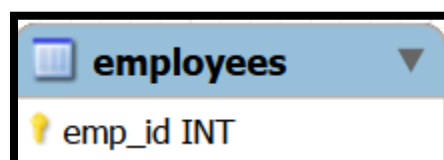
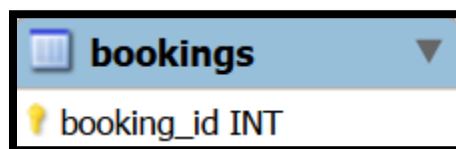
30. Primary Keys

Primary keys are used to uniquely identify each record in a table. A primary key value cannot be duplicated and cannot be null. This makes it possible to distinguish one record from another.

In this project, most tables have an ID column as the primary key. For example, the employees table uses emp_id, the hajj_applicants table uses hajji_id, and the bookings table uses booking_id.

Primary keys are important because they are used by other tables to create relationships. For example, the emp_id in the employees table can be referenced by the employee_tasks table to connect tasks to employees.

Using primary keys helps keep the database organized and prevents duplicate identifiers.



31. Foreign Keys

Foreign keys are used to connect one table to another. A foreign key in one table refers to a primary key in another table. This ensures that related records are connected correctly.

For example, the `employee_tasks` table contains `emp_id` as a foreign key. This value refers to the `emp_id` in the `employees` table. This means that a task cannot be assigned to an employee unless that employee exists in the `employees` table.

Another example is the `bookings` table. It contains `hajji_id` and `package_id`. The `hajji_id` connects the booking to a Hajj applicant, and `package_id` connects the booking to a package.

Foreign keys help protect the database from invalid relationships. They prevent records from referring to data that does not exist.

Foreign keys

```
'ER TABLE `bookings` ADD CONSTRAINT `bookings_hajji_id_fkey` FOREIGN KEY (`hajji_id`) REFERENCES `hajj_applicants` (`hajji_id`) ON DELETE CASCADE;
'ER TABLE `bookings` ADD CONSTRAINT `bookings_package_id_fkey` FOREIGN KEY (`package_id`) REFERENCES `packages` (`package_id`) ON DELETE SET NULL;
'ER TABLE `employee_messages` ADD CONSTRAINT `employee_messages_emp_id_fkey` FOREIGN KEY (`emp_id`) REFERENCES `employees` (`emp_id`) ON DELETE CASCADE;
'ER TABLE `employee_tasks` ADD CONSTRAINT `employee_tasks_emp_id_fkey` FOREIGN KEY (`emp_id`) REFERENCES `employees` (`emp_id`) ON DELETE CASCADE;
'ER TABLE `hajj_applicants` ADD CONSTRAINT `hajj_applicants_country_id_fkey` FOREIGN KEY (`country_id`) REFERENCES `country` (`country_id`) ON DELETE SET NULL;
'ER TABLE `hajj_applicants` ADD CONSTRAINT `hajj_applicants_supervisor_id_fkey` FOREIGN KEY (`supervisor_id`) REFERENCES `supervisor` (`supervisor_id`) ON DELETE SET NULL;
'ER TABLE `hajj_applicants` ADD CONSTRAINT `hajj_applicants_user_id_fkey` FOREIGN KEY (`user_id`) REFERENCES `users` (`id`) ON DELETE SET NULL;
'ER TABLE `hajji_transports` ADD CONSTRAINT `hajji_transports_hajji_id_fkey` FOREIGN KEY (`hajji_id`) REFERENCES `hajj_applicants` (`hajji_id`) ON DELETE CASCADE;
'ER TABLE `hajji_transports` ADD CONSTRAINT `hajji_transports_transport_id_fkey` FOREIGN KEY (`transport_id`) REFERENCES `transports` (`transport_id`) ON DELETE CASCADE;
'ER TABLE `hotel_bookings` ADD CONSTRAINT `hotel_bookings_hajji_id_fkey` FOREIGN KEY (`hajji_id`) REFERENCES `hajj_applicants` (`hajji_id`) ON DELETE CASCADE;
'ER TABLE `hotel_bookings` ADD CONSTRAINT `hotel_bookings_hotel_id_fkey` FOREIGN KEY (`hotel_id`) REFERENCES `hotels` (`hotel_id`) ON DELETE SET NULL;
'ER TABLE `transports` ADD CONSTRAINT `transports_company_id_fkey` FOREIGN KEY (`company_id`) REFERENCES `transport_companies` (`company_id`) ON DELETE SET NULL;
```

32. Composite Primary Key Example

Some tables may use a composite primary key. A composite primary key is a primary key made of more than one column. This is used when one column alone is not enough to uniquely identify a record.

In this project, the `hajji_transports` table can use a composite key made of `hajji_id` and `transport_id`. This means that the same Hajj applicant cannot be connected to the same transport trip more than once.

For example, the pair (`hajji_id` = 1, `transport_id` = 2) must be unique. However, the same Hajj applicant can be connected to another transport trip, and the same transport trip can be connected to another Hajj applicant.

When deleting a record from a table with a composite key, both columns are usually used in the WHERE condition. For example:

```
DELETE FROM hajji_transports WHERE hajji_id = 1 AND transport_id = 2;
```

	hajji	hajji	hajji
	1	1	caucsej6q
	hajji_id	hajji_id	hajji_id

hajji_transports

33. Constraints and Data Integrity

Constraints are rules applied to database tables to protect the data and keep it valid. They prevent incorrect data from being stored and help maintain relationships between tables.

The system uses several types of constraints. Primary keys prevent duplicate IDs. Foreign keys maintain relationships between tables. NOT NULL prevents important fields from being empty. UNIQUE prevents duplicate values such as usernames. CHECK restricts values to a valid range, such as hotel rating values. AUTO_INCREMENT automatically generates ID values.

Data integrity is very important in this project because many tables depend on each other. For example, a booking must be linked to an existing Hajj applicant and an existing package. A task must be linked to an existing employee. A hotel booking must be linked to an existing hotel and applicant.

These constraints make the database more reliable and reduce data entry errors.

```
CONSTRAINT `hotels_pkey` PRIMARY KEY (`hotel_id`)
```

PRIMARY KEY

```
ALTER TABLE `bookings` ADD CONSTRAINT `bookings_hajji_id_fkey` FOREIGN KEY (`hajji_id`) REFERENCES `hajj_applicants` (`hajji_id`) ON DELETE CASCADE;
```

FOREIGN KEY

```
CONSTRAINT `transport_companies_rating_check` CHECK (`rating` >= 1 AND `rating` <= 5),
```

CHECK

```
`company_id` INT AUTO_INCREMENT NOT NULL,
```

AUTO_INCREMENT

34. Constraint Types Used in the System

Constraint Type	Purpose	Example in the System
Primary Key	Uniquely identifies records.	emp_id, hajji_id, booking_id
Foreign Key	Connects tables together.	employee_tasks.emp_id references employees.emp_id
Not Null	Prevents empty required fields.	Required names, roles, or IDs
Unique	Prevents duplicate values.	Unique username in users
Check	Limits values to valid options or ranges.	Rating between 1 and 5
Auto Increment	Automatically generates numeric IDs.	Campaign ID or employee ID

35. ON DELETE Actions

The database can use special actions with foreign keys, such as ON DELETE CASCADE and ON DELETE SET NULL. These actions define what happens to child records when a parent record is deleted.

ON DELETE CASCADE means that when the parent record is deleted, related child records are also deleted automatically. For example, if a Hajj applicant is deleted, related booking records may be deleted depending on the relationship design.

ON DELETE SET NULL means that when the parent record is deleted, the child record remains, but the foreign key value becomes NULL. This is useful when the system should keep the child record but remove the link to the deleted parent record.

These actions help maintain database consistency and reduce manual work when deleting records.

```
ON DELETE CASCADE;  
ON DELETE SET NULL;
```

36. Normalization

Normalization is the process of organizing database tables to reduce data duplication and improve data consistency. It helps divide data into logical tables and connect them using relationships.

In this project, normalization was applied by separating the data into different tables. For example, employee information is stored in the employees table, while employee tasks are stored in employee_tasks. This prevents repeating employee information for every task.

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Also, package details are stored in the packages table, while customer bookings are stored in the bookings table. This means that package information does not need to be repeated inside every booking record.

Hotel information is stored in the hotels table, while customer hotel reservations are stored in hotel_bookings. Transport company information is stored in transport_companies, while transport trips are stored in transports.

This design makes the database easier to update. If hotel information changes, it can be updated once in the hotels table instead of changing many reservation records.

37. First Normal Form

The database follows the First Normal Form because each column contains atomic values. This means that each field stores one value only and does not contain multiple values in the same cell.

For example, in the employees table, employee first name, last name, email, gender, and salary are stored in separate columns. In the hotels table, hotel name, phone, city, state, and rating are also stored in separate columns.

This makes the data easier to search, update, and display in the application interfaces.

	emp_id	first_name	last_name	email	gender	salary
▶	1	bb	noor	ahmad@example.com	Male	700.00
	2	ameen	emad	ameen@gmail.com	male	100.00
*	NULL	NULL	NULL	NULL	NULL	NULL

employees

	hotel_id	hotel_name	phone	street_address	zip	city	state	rating
▶	1	Makkah Hotel	123456789	Central Area	24231	Makkah	Makkah Region	5
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

hotels

38. Second Normal Form

The database follows the Second Normal Form because non-key attributes depend on the primary key of their table. Each table stores information related to its own main entity.

For example, in the employees table, employee name, email, gender, and salary depend on emp_id. In the packages table, package name, start date, end date, and price depend on package_id.

The system avoids storing unrelated information in the same table. Employee tasks are not stored inside the employees table. Instead, they are stored in employee_tasks and connected to employees using emp_id.

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	emp_id	first_name	last_name	email	gender	salary
▶	1	bb	noor	ahmad@example.com	Male	700.00
	2	ameen	emad	ameen@gmail.com	male	100.00
*	NULL	NULL	NULL	NULL	NULL	NULL

employees

	task_id	emp_id	title	description	status	due_date
▶	1	1	Check bookings	Review customer bookings for today	Done	2026-05-11
	2	1	Review hotel bookings	Check hotel reservations and room types	Done	2026-05-21
*	NULL	NULL	NULL	NULL	NULL	NULL

employee_tasks

39. Third Normal Form

The database follows the Third Normal Form because it separates data that depends on other non-key information into different tables. This prevents indirect dependencies and reduces repeated data.

For example, hotel information is stored in the hotels table, while hotel reservation information is stored in hotel_bookings. The reservation table stores hotel_id instead of repeating hotel name, city, phone, and rating for every reservation.

Also, transport company information is stored in transport_companies, while transport trip information is stored in transports. This prevents repeating company information inside every transport trip.

This structure makes updates easier and helps maintain consistent data across the system.

	hotel_id	hotel_name	phone	street_address	zip	city	state	rating
▶	1	Makkah Hotel	123456789	Central Area	24231	Makkah	Makkah Region	5
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

hotels

	booking_id	hajji_id	hotel_id	room_type	check_in	check_out	status	price
▶	1	1	1	Double	2026-06-01	2026-06-10	cancelled	800.00
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

hotel_bookings

	transport_id	company_id	vehide_type	from_location	to_location	departure_date
▶	1	1	Bus	Jeddah Airport	Makkah	2026-06-01
	2	1	plane	palestine	makkah	2026-06-05
*	NULL	NULL	NULL	NULL	NULL	NULL

transports

	company_id	company_name	email	phone	rating
▶	1	Hajj Transport Co	transport@example.com	555555555	4
*	NULL	NULL	NULL	NULL	NULL

transport_companies

40. SQL Implementation

SQL was used to create and manage the database of the system. The SQL script includes commands for creating tables, defining columns, setting data types, adding primary keys, adding foreign keys, and inserting sample data.

The script creates each table separately based on its purpose in the system. For example, the users table is created to store login accounts, the employees table is created to store employee information, and the bookings table is created to store customer booking records.

SQL is also used to apply database constraints. These constraints make sure that the data is valid and consistent. For example, primary keys prevent duplicate IDs, while foreign keys ensure that records are connected to existing records in related tables.

The SQL implementation is important because it represents the structure of the whole database. Without the SQL script, the application would not be able to store, retrieve, update, or delete data correctly.

```
CREATE TABLE `campaigns` (  
  `id` INT AUTO_INCREMENT NOT NULL,  
  `name` VARCHAR(100) NOT NULL,  
  `location` VARCHAR(100),  
  `email` VARCHAR(100),  
  CONSTRAINT `campaigns_pkey` PRIMARY KEY (`id`)  
)
```

Figure: SQL script used to create database tables.

41. Creating Tables Using SQL

The database tables were created using the CREATE TABLE command. This command defines the table name, column names, data types, and constraints.

Each table was created according to the type of data it stores. For example, text values such as names and emails are stored using VARCHAR, dates are stored using DATE, and prices or salaries are stored using numeric types such as DECIMAL.

The use of correct data types is important because it helps the database store data properly. For example, storing dates as DATE allows the system to sort and compare dates correctly. Storing prices as numeric values allows the system to calculate and display prices correctly.

Creating tables properly is the first step in building a reliable database system.

```
CREATE TABLE `bookings` (  
  `booking_id` INT AUTO_INCREMENT NOT NULL,  
  `hajji_id` INT,  
  `package_id` INT,  
  `status` VARCHAR(30) DEFAULT 'Pending',  
  `total_price` DECIMAL(10,2) DEFAULT 0,  
  CONSTRAINT `bookings_pkey` PRIMARY KEY (`booking_id`)  
)
```

```
CREATE TABLE `hajj_applicants` (  
  `hajj_id` INT AUTO_INCREMENT NOT NULL,  
  `user_id` INT,  
  `first_name` VARCHAR(50) NOT NULL,  
  `last_name` VARCHAR(50) NOT NULL,  
  `gender` VARCHAR(20),  
  `phone` VARCHAR(30),  
  `dob` DATE,  
  `status` VARCHAR(30) DEFAULT 'Pending',  
  `supervisor_id` INT,  
  `country_id` INT,  
  CONSTRAINT `hajj_applicants_pkey` PRIMARY KEY (`hajj_id`)  
)
```

42. Insert Operations

The INSERT command is used to add new records to database tables. In this system, insert operations are used when the Manager adds new campaigns, employees, hotels, or transport companies. Insert operations are also used to add sample data such as users, packages, bookings, tasks, and messages.

For example, when a new employee is added from the Employees Data Management interface, the Java application sends an INSERT query to the database. After the record is inserted, the table in the interface is refreshed to display the new data.

Using insert operations allows the system to grow and store new records as the company continues to work.

```
• INSERT INTO `users` (`id`, `username`, `password`, `role`) VALUES (1, 'admin', 'admin123', 'admin');  
• INSERT INTO `users` (`id`, `username`, `password`, `role`) VALUES (2, 'employee', 'emp123', 'employee');  
• INSERT INTO `users` (`id`, `username`, `password`, `role`) VALUES (3, 'customer', 'cust123', 'customer');
```

43. Select Operations

The SELECT command is used to retrieve data from the database. It is one of the most important SQL commands in the system because most interfaces need to display data inside tables.

For example, the Manager Dashboard uses SELECT queries to show campaigns, employees, Hajj applicants, hotels, and transport companies. The Employee Dashboard uses SELECT queries to display the employee profile, assigned tasks, and messages. The Customer Dashboard uses SELECT queries to show booking, hotel, and transport information.

Some SELECT queries use joins to combine data from multiple tables. For example, the Customer Booking interface can display the package name from the packages table and the booking status from the bookings table.

This makes the displayed data more useful and easier to understand for the user.

```
SELECT * FROM bookings
```

	booking_id	hajji_id	package_id	status	total_price
▶	1	1	1	cancelled	2500.00
*	NULL	NULL	NULL	NULL	NULL

44. Update Operations

The UPDATE command is used to modify existing records in the database. In this system, update operations are used in many parts of the application.

For example, the Manager can update campaign information, employee records, Hajj applicant status, hotel information, and transport company details. The Employee can update task status from Pending to Done. The Customer can update limited information such as phone number or password.

Update operations are important because company data changes over time. For example, an applicant status may change from Pending to Approved, a task may change from Pending to Done, or a booking may change to Cancelled.

After an update operation, the interface refreshes the displayed data so the user can see the latest information.

	task_id	emp_id	title	description	status	due_date
▶	1	1	Check bookings	Review customer bookings for today	Done	2026-05-11
	2	1	Review hotel bookings	Check hotel reservations and room types	Done	2026-05-21
	3	1	a	b	Pending	2020-05-25
✱	NULL	NULL	NULL	NULL	NULL	NULL

insert into employee_tasks values (3 , 1 , 'a' , 'b' , 'Pending' , '2020-05-25')

45. Delete Operations

The DELETE command is used to remove records from the database. In the Manager Dashboard, delete operations are available in management screens such as Campaigns Management, Employees Data Management, Hotel Management, and Transport Company Management.

Deleting records should be done carefully because some records may be connected to other tables using foreign keys. For this reason, the database uses relationships and constraints to protect data integrity.

In some customer-related screens, the system may not delete records completely. Instead, it may update the status to Cancelled. This is useful because it keeps the history of the record while showing that the service is no longer active.

For example, when a customer cancels a booking, the booking record can remain in the database with status Cancelled. This helps the company keep a record of cancelled services.

	booking_id	hajji_id	package_id	status	total_price
▶	1	1	1	cancelled	2500.00
✱	NULL	NULL	NULL	NULL	NULL

status = Cancelled

46. SQL CRUD Summary Table

SQL Operation	Purpose	Example in the System
INSERT	Adds new records.	Add employee, campaign, hotel, or task.
SELECT	Retrieves and displays data.	Display records in JTable.
UPDATE	Modifies existing records.	Update status, phone number, or employee data.
DELETE	Removes records.	Delete campaign or employee record.

47. Sample Data in the Database

Sample data was inserted into the database to test the system and make sure that all interfaces work correctly. The sample data includes users, employees, Hajj applicants, campaigns, packages, bookings, hotels, transport companies, tasks, and messages.

The sample users were used to test the login system and verify that each role opens the correct dashboard. Employee sample data was used to test the Employee Dashboard, including profile, tasks, and messages. Customer sample data was used to test booking, hotel, and transport screens.

Sample data is important because it allows testing the system before real data is used. It helps confirm that database relationships are correct and that the Java interfaces can retrieve and display records successfully.

48. Sample Data Examples

Examples of sample data used in the system include:

Table	Example Data
Users	Admin, Employee, and Customer login accounts.
Employees	Employee names, emails, gender, and salary.
hajj_applicants	Customer names, phone numbers, gender, and application status.
Packages	Standard Hajj Package with start date, end date, and price.
Bookings	Customer booking connected to a Hajj package.
Hotels	Hotel name, city, phone, and rating.
employee_tasks	Tasks such as checking bookings or preparing reports.
employee_messages	Internal messages between management and employees.

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49. Importance of Sample Data

Sample data helped in testing the connection between the graphical user interfaces and the database. When records appeared correctly in the interfaces, this confirmed that the SELECT queries were working. When new records were added from the interfaces and appeared in MySQL Workbench, this confirmed that the INSERT queries were working.

Sample data also helped test update operations. For example, changing task status from Pending to Done confirmed that the UPDATE query worked correctly. Cancelling a booking and seeing the status change to Cancelled confirmed that the system can track service changes.

This testing process helped verify that the system is functioning correctly and that the database is connected properly with the Java application.

Before :

	task_id	emp_id	title	description	status	due_date
▶	1	1	Check bookings	Review customer bookings for today	Done	2026-05-11
	2	1	Review hotel bookings	Check hotel reservations and room types	Done	2026-05-21
	3	1	a	b	Pending	2020-05-25
*	NULL	NULL	NULL	NULL	NULL	NULL

After :



✔ Done Tasks: 2

✖ Pending Tasks: 1

My Tasks

Status:

🔄 Refresh

✅ Mark as Done

🚪 Logout

Employee can only view and update task status

My Information

Task ID	Title	Description	Status	Due Date
3	a	b	Pending	2020-05-25
1	Check bookings	Review customer bookings for tod...	Done	2026-05-11
2	Review hotel bookings	Check hotel reservations and roo...	Done	2026-05-21

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Done Tasks: 3

Pending Tasks: 0

Status: D...

Refresh

Mark as Done

Employee can only view and update task status

Logout

My Information

Task ID	Title	Description	Status	Due Date
3	a	b	Done	2020-05-25
1	Check bookings	Review customer bookings for tod...	Done	2026-05-11
2	Review hotel bookings	Check hotel reservations and roo...	Done	2026-05-21

	task_id	emp_id	title	description	status	due_date
▶	1	1	Check bookings	Review customer bookings for today	Done	2026-05-11
	2	1	Review hotel bookings	Check hotel reservations and room types	Done	2026-05-21
	3	1	a	b	Done	2020-05-25
*	NULL	NULL	NULL	NULL	NULL	NULL

50. Java Implementation

The application was implemented using Java in NetBeans. Java was used to build the logic of the system, while Java Swing was used to design the graphical user interfaces. The project contains several forms and panels, and each one is responsible for a specific part of the system.

The Java application is divided according to the system users. There are interfaces for the Manager, Employee, and Customer. Each interface contains buttons, text fields, combo boxes, labels, and tables that allow the user to interact with the database.

For example, the Manager interfaces allow the Manager to add, update, delete, and view records. The Employee interfaces allow employees to view tasks and messages. The Customer interfaces allow customers to view booking, hotel, and transport information.

This structure makes the application easier to understand and maintain because each screen has a clear purpose.

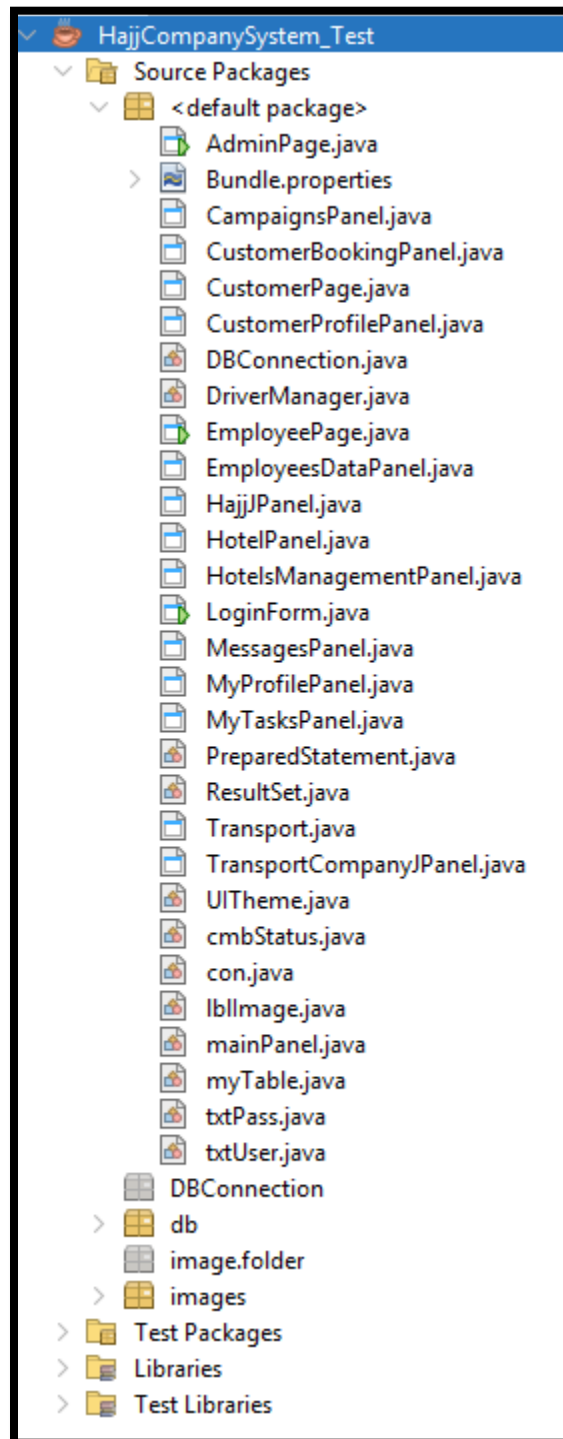


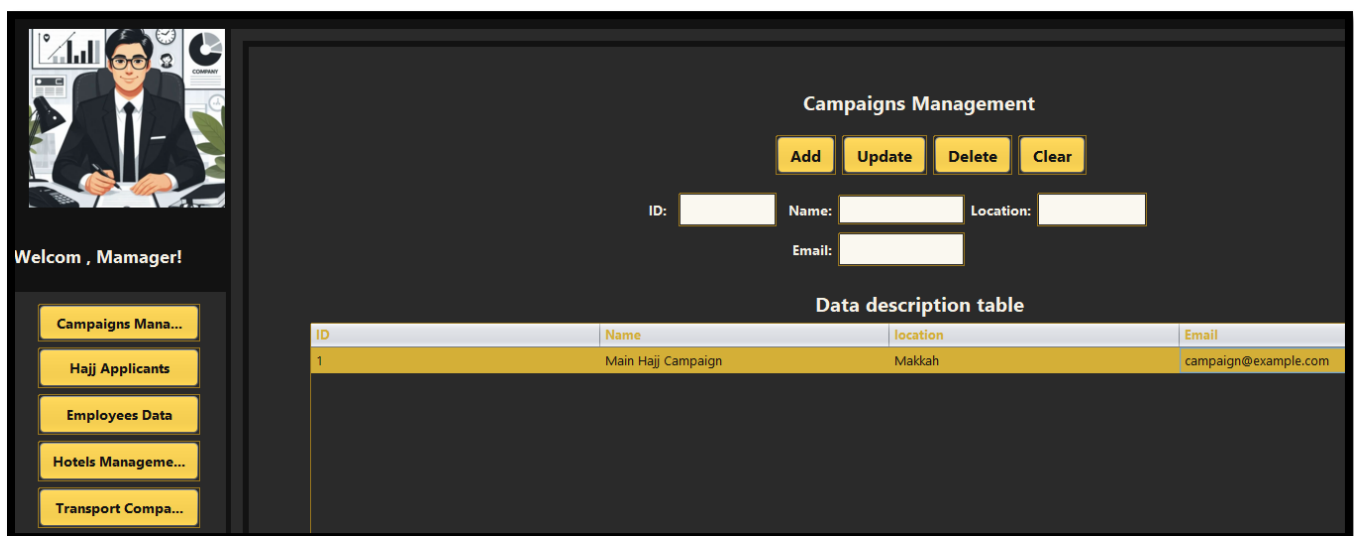
Figure: Java project structure in NetBeans.

51. Java Swing GUI

Java Swing was used to create the graphical user interface of the system. Swing provides components such as JFrame, JPanel, JButton, JLabel, JTextField, JComboBox, and JTable. These components were used to build the application screens.

The system uses buttons to perform actions such as adding, updating, deleting, refreshing, and logging out. Text fields are used to enter data, combo boxes are used to select values such as gender or status, and tables are used to display records from the database.

The GUI was designed using a dark and gold color theme to make the application look more professional and suitable for a Hajj and Umrah company system. The side menus also make navigation easier for users.



52. JDBC Connection

JDBC stands for Java Database Connectivity. It is used to connect the Java application with the MySQL database. Without JDBC, the Java application would not be able to send SQL queries to the database or retrieve data from it.

In this project, JDBC is responsible for opening the connection between NetBeans and MySQL. After the connection is created, the application can execute SQL commands such as INSERT, SELECT, UPDATE, and DELETE.

The general flow of database interaction is:

User Action → Java Code → JDBC → MySQL Database → Result → GUI Display

For example, when the Manager clicks the Add button, Java uses JDBC to send an INSERT query to MySQL. After the data is saved, the table in the interface is refreshed and the new record appears.



53. DBConnection Class

The database connection is handled in a class called DBConnection.java. This class contains the database URL, username, and password. It also contains the method that opens the connection with the MySQL database.

The connection URL includes the database location, port number, and database name. For example, the URL may connect to a local MySQL database running on port 3306. When the application starts or when a screen needs database access, it calls the connection method from this class.

Using a separate connection class is useful because it keeps the database connection code in one place. This makes the project easier to maintain. If the database name, username, or password changes, the developer can update the connection information in one class instead of changing it in many screens.

```
public class DBConnection {  
  
    private static final String URL = "jdbc:mysql://localhost:3306/hajj_company?useSSL=false&serverTimezone=UTC";  
    private static final String USER = "root";  
    private static final String PASS = "12324013";  
  
    public static Connection connect() {  
        try {  
            Class.forName("com.mysql.cj.jdbc.Driver");  
            return DriverManager.getConnection(URL, USER, PASS);  
        } catch (Exception e) {  
            System.out.println("Connection Error: " + e.getMessage());  
            return null;  
        }  
    }  
}
```

Figure: DBConnection class used to connect Java with MySQL.

54. PreparedStatement Usage

The project uses PreparedStatement to execute SQL queries from Java. A PreparedStatement is used to write SQL commands with placeholders, then fill these placeholders with values from the interface.

Using PreparedStatement makes the code cleaner and safer. It helps avoid writing user input directly inside SQL strings. It is used in many operations such as adding records, updating records, deleting records, and selecting records.

For example, when the Manager adds a new campaign, the values entered in the text fields are passed to a PreparedStatement. Then the query is executed and the new record is saved in the database.

This approach also makes the code easier to read because the SQL query and the inserted values are separated clearly.

```
try {  
    String sql =  
        "UPDATE bookings SET status='cancelled' WHERE booking_id=?";  
    PreparedStatement pst = con.prepareStatement(sql);  
    pst.setInt(1, bookingId);  
    pst.executeUpdate();  
  
    JOptionPane.showMessageDialog(this,  
        "Booking cancelled successfully");  
    loadMyBookings();  
} catch (Exception e) {
```

55. ResultSet Usage

The project uses ResultSet to store data returned from SELECT queries. When the application sends a SELECT query to MySQL, the database returns records. These records are stored temporarily in a ResultSet.

The Java code then reads the data from the ResultSet row by row. Each value is extracted using methods such as getString, getInt, getDate, or getDouble, depending on the data type.

For example, when the Employee opens the My Tasks screen, the application sends a SELECT query to retrieve the employee tasks. The returned tasks are stored in a ResultSet, then added to the table in the interface.

This allows the application to display database records inside the graphical user interface.

Hajj Report

```
int updated = pst.executeUpdate();
if (updated > 0) {
    JOptionPane.showMessageDialog(this, "Task marked as Done!");
    loadTasks();
} else {
    JOptionPane.showMessageDialog(this, "No task updated (check task belongs to you).");
}
```

56. JTable Display

The system uses JTable to display database records inside the graphical user interface. Tables are used in many screens, such as Campaigns Management, Hajj Applicants Management, Employees Data Management, My Tasks, My Messages, My Booking, My Transport, and My Hotel.

When data is retrieved from the database using a SELECT query, it is added to a table model such as DefaultTableModel. The table model controls the rows and columns displayed in the JTable.

Before loading new data, the system usually clears the old rows using `model.setRowCount(0)`. Then it adds the new records row by row. This ensures that the table displays updated data without duplication.

Using JTable makes the system easier to use because users can see records in an organized table format.

The screenshot shows a dark-themed window titled "Hajj Management". At the top, there are four yellow buttons: "Add", "Update", "Delete", and "Clear". Below these buttons is a form with several input fields and dropdown menus. The fields are labeled "ID:", "First:", "Last:", "Gender:", "DOB:", "Phone:", and "Status:". The "Gender" dropdown is currently set to "M..." and the "Status" dropdown is currently set to "Pending".

57. General Application Workflow

The application workflow starts when the user opens the system and logs in. The login form checks the username and password from the database. If the information is correct, the system checks the user role and opens the correct dashboard.

After the dashboard opens, the user can interact with the available screens. The Manager can use management screens to add, update, delete, and view records. Employees can view tasks and messages. Customers can view booking, hotel, and transport information.

Hajj Report

Every button click in the interface is connected to Java code. The Java code sends SQL queries to the database through JDBC. The database processes the query and returns a result. The result is then displayed in the GUI.

This workflow connects the user interface, Java code, JDBC, and MySQL database into one complete system.

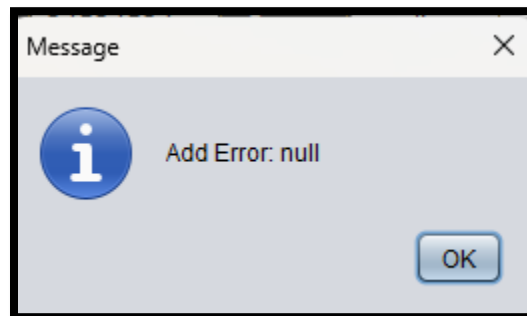
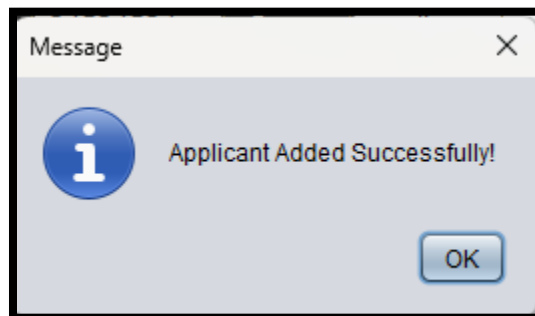
Login → Dashboard → Button Click → SQL Query → Database → JTable Display

58. Error Handling in the Application

The application uses message dialogs to notify the user about successful operations or errors. For example, when a record is added successfully, the system displays a success message. If an error occurs, the system displays an error message explaining the problem.

Error handling is important because it helps the user understand what happened. For example, if required fields are empty, the system can show a warning message. If a database error occurs, the system can show the error message returned from the database.

Using JOptionPane helps display these messages in a simple and user-friendly way. This makes the application easier to test and use.



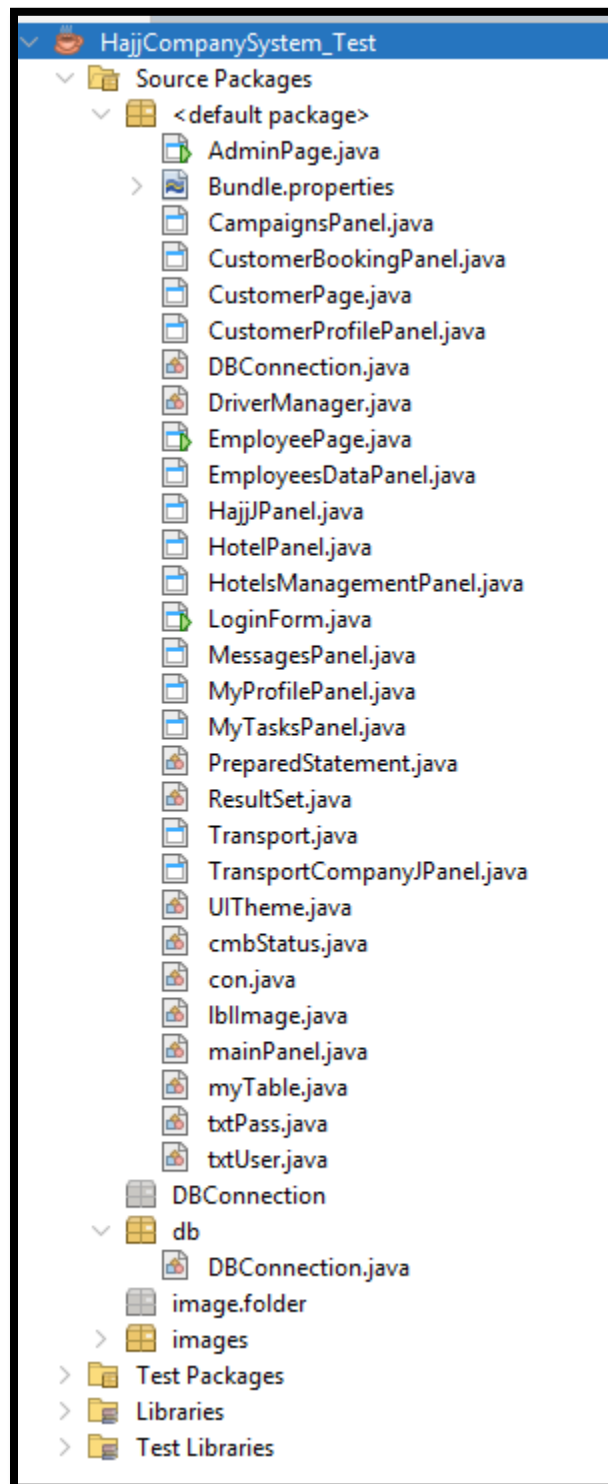
59. Code Organization

The project code is organized into different classes and panels. Each class is responsible for a specific part of the system. For example, the login class handles user authentication, the manager panels handle management screens, the employee panels handle employee services, and the customer panels handle customer services.

Hajj Report

This organization makes the project easier to understand and modify. If the developer wants to update the employee tasks screen, they can open the related panel only. If they want to update the database connection, they can modify the DBConnection.java class.

Organizing the code into separate classes also improves maintainability and makes future development easier.



60. Login Interface

The Login Interface is the first screen that appears when the system starts. It allows users to enter their username and password. The system then checks the entered information from the users table in the database.

After the username and password are verified, the system checks the user role. Based on the role, it opens the correct dashboard. If the user is a Manager, the Manager Dashboard opens. If the user is an Employee, the Employee Dashboard opens. If the user is a Customer, the Customer Dashboard opens.

This interface is important because it controls access to the system. It prevents users from opening dashboards that are not related to their role. It also gives the system a clear starting point.

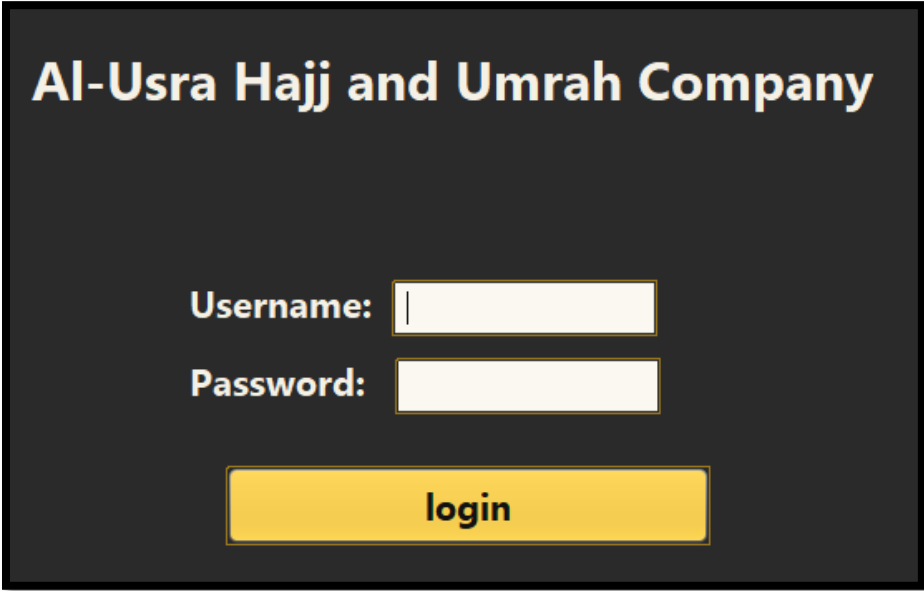
The image shows a login interface for 'Al-Usra Hajj and Umrah Company'. The title is at the top in a bold, yellow font. Below it, there are two input fields: one for 'Username:' and one for 'Password:'. Both fields are white with a yellow border. Below the password field is a yellow button with the word 'login' in black text.

Figure: Login interface used for role-based access.

61. Manager Dashboard Overview

The Manager Dashboard is the main interface for the Manager role. It gives the Manager access to the main management sections of the system. These sections include Campaigns Management, Hajj Applicants Management, Employees Data Management, Hotels Management, and Transport Companies Management.

The dashboard uses a side menu to make navigation easier. The Manager can click on any section to open its related management screen. This design helps organize the Manager functions in one place and makes the system easier to use.

The Manager Dashboard is connected to many database tables because the Manager is responsible for managing the main company data. Most CRUD operations are performed from this dashboard.

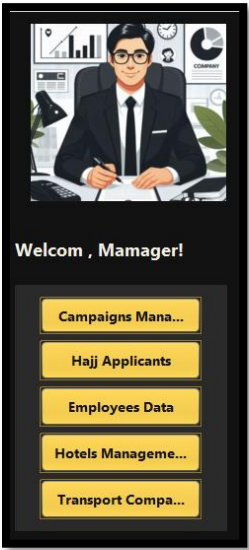


Figure: Manager Dashboard main navigation.

62. Campaigns Management Interface

The Campaigns Management interface allows the Manager to manage Hajj campaign records. The interface contains fields for campaign ID, campaign name, location, and email. It also contains buttons for adding, updating, deleting, and clearing data.

When the Manager adds a new campaign, the data is inserted into the campaigns table. When the Manager selects an existing campaign, the information can be updated or deleted. The table at the bottom displays campaign records stored in the database.

This interface is a clear example of CRUD operations in the system. It allows the Manager to create, read, update, and delete campaign records directly from the graphical interface.

Campaigns Management

Add

Update

Delete

Clear

ID:

Name:

Location:

Email:

Data description table

ID	Name	location	Email
1	Main Hajj Campaign	Makkah	campaign@example.com

Figure: Campaigns Management interface showing CRUD operations.

63. Hajj Applicants Management Interface

The Hajj Applicants Management interface allows the Manager to manage pilgrims or customers who applied for Hajj services. The interface includes fields such as ID, first name, last name, gender, date of birth, phone number, and status.

The status field is important because it helps track the progress of each applicant. The status can be Pending, Approved, Rejected, Cancelled, or Completed. This allows the company to know the current state of each Hajj application.

The table in this interface displays applicant records from the hajj_applicants table. The Manager can add new applicants, update existing information, delete records, or clear the input fields.

This interface is connected to the customer side of the system because each customer is represented as a Hajj applicant in the database.

Hajjaj Management

Add **Update** **Delete** **Clear**

ID: First: Last: Gender: M...

DOB: Phone: Status: Pending

Data description table

ID	First Name	Last Name	Phone	status	Gender	DateOfBirth
1	mohammed	jaber	45454545	Pending	Male	2000-01-01
2	ali	ameen	056243	Pending	Male	2000-01-08
10	v	b	34534534	Pending	Male	2000-01-01

Figure: Hajj Applicants Management interface.

64. Employees Data Management Interface

The Employees Data Management interface allows the Manager to manage employee records. The interface contains fields such as employee ID, first name, last name, salary, email, and gender. It also includes buttons for adding, updating, deleting, and clearing employee data.

The employee records are stored in the employees table. The Manager can add new employees, update employee information, delete employee records, and view employee data in a table.

This interface also includes an internal messaging section. The Manager can send messages to employees, and these messages are stored in the employee_messages table. Employees can later view these messages in their dashboard.

Hajj Report

This interface is important because it combines employee data management and communication between management and employees.

Employee data management

Add Update Delete Clear

ID: First: Last: Salary:

Email: Gender: Male

Data description table

ID	First Name	Last Name	Email	Gender	Salary
1	bb	noor	ahmad@example.com	Male	700.00
2	ameen	emad	ameen@gmail.com	male	100.00

I implemented internal messaging between admin and employees

Send Message

Send

Figure: Employees Data Management and internal messaging section.

65. Hotels Management Interface

The Hotels Management interface allows the Manager to manage hotel records used by the company. The interface contains fields such as hotel ID, hotel name, phone number, rating, city, state, street address, and ZIP code.

The Manager can add new hotels, update hotel information, delete hotel records, and view all hotels in a table. The hotel data is stored in the hotels table.

This interface is important because hotel information is used later in the Customer Hotel interface. A hotel must exist in the hotels table before it can be connected to a customer hotel booking.

The interface helps the company organize hotel service providers and keep hotel information updated.

Hotel Management

AddUpdateDeleteClear

ID: Name: Phone:

City: State: Street: ZIP:

Hotel description table

ID	Hotel Name	Phone	Street Address	ZIP	City	State	Rating
1	Makkah Hotel	123456789	Central Area	24231	Makkah	Makkah Region	5

Figure: Hotel Management interface.

66. Transport Company Management Interface

The Transport Company Management interface allows the Manager to manage transportation company records. The interface contains fields such as company ID, company name, phone number, email, and rating.

The data entered in this interface is stored in the transport_companies table. The Manager can add new transport companies, update company information, delete companies, and view existing records.

This interface is important because transport companies are connected to transport trips. A transport company can provide many transport trips, and those trips can later be assigned to Hajj applicants.

By managing transport companies separately, the system avoids repeating company information inside every transport record.

Transport Company Management

AddUpdateDeleteClear

ID: Name: Phone: Rating:

Email:

Hotel description table

ID	Company_name	Email	Phone	Rating
1	Hajj Transport Co	transport@example.com	555555555	4

Figure: Transport Company Management interface.

67. Manager Dashboard Summary

The Manager Dashboard is the most complete dashboard in the system because it allows the Manager to control the main company records. The Manager can manage campaigns, Hajj applicants, employees, hotels, and transport companies. These screens are directly connected to the main database tables.

The Manager Dashboard supports CRUD operations and provides tables to display records. It also supports communication with employees through the messaging feature.

This dashboard is essential because it provides the administrative side of the system and keeps the company data organized.

68. Employee Dashboard Overview

The Employee Dashboard is designed for employees who work inside the company. It provides a simple interface that allows employees to access their personal information, assigned tasks, and internal messages.

The Employee Dashboard contains a side menu that helps the employee move between the main sections. These sections include My Profile, My Tasks, Messages, and Logout. This makes the dashboard easy to use and keeps employee functions organized.

Unlike the Manager Dashboard, the Employee Dashboard does not allow the employee to manage all company records. The employee has limited permissions and can only access information related to their work. This supports the role-based access concept used in the system.

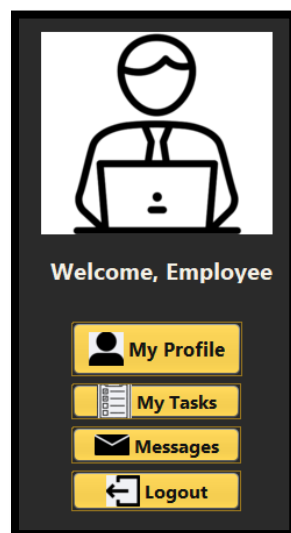


Figure: Employee Dashboard navigation menu.

69. Employee My Profile Interface

The My Profile interface allows the employee to view personal information stored in the database. This information includes employee ID, first name, last name, email, gender, and salary.

The interface displays the employee information in a table and also provides fields for updating some personal information. When the employee updates the profile, the system sends an UPDATE query to the employees table.

This interface is connected mainly to the employees table. It helps employees review their information and ensures that their profile data is available in an organized way.

The interface may also include a Sort by Salary button. This button can sort employee records by salary. In the project, sorting is done using an SQL query with ORDER BY salary DESC, which means salaries are displayed from highest to lowest.

Employee_id	first_name	last_name	Email	Gender	Salary
1	bb	noor	ahmad@example.com	Male	700.0

Figure: Employee profile interface displaying employee information.

70. Employee Tasks Interface

The My Tasks interface allows employees to view the tasks assigned to them. Each task contains a task ID, title, description, status, and due date. The task data is retrieved from the employee_tasks table.

This interface displays only the tasks related to the logged-in employee. This is done by filtering tasks using emp_id. For example, if the logged-in employee has emp_id = 2, the interface should display only tasks assigned to employee number 2.

The interface also includes task counters, such as the number of pending tasks and the number of completed tasks. These counters help employees understand their work progress quickly.

The employee can select a task and click Mark as Done. When this button is clicked, the system updates the task status from Pending to Done in the database.

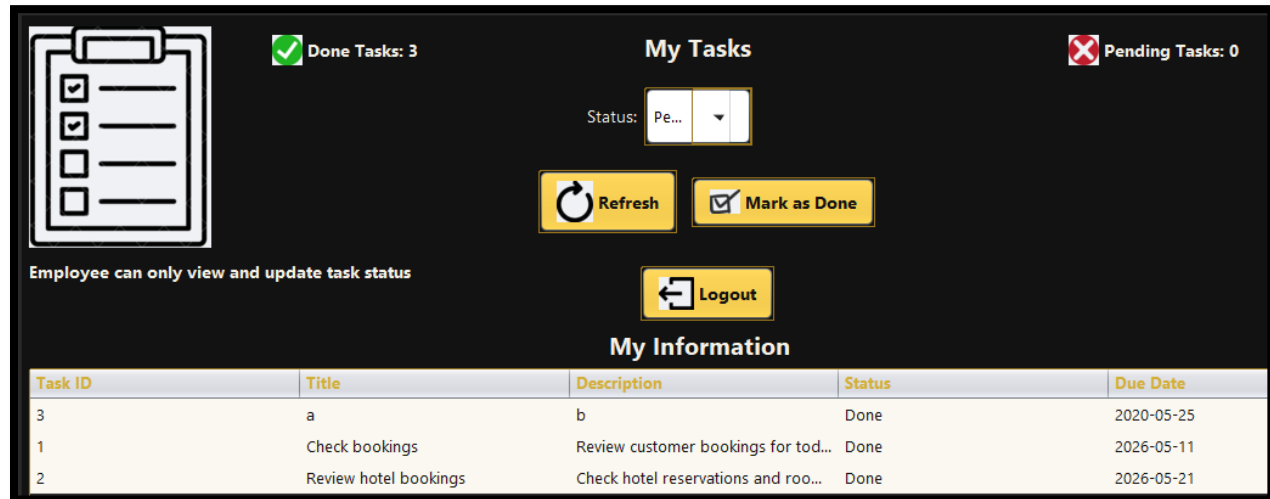


Figure: Employee tasks interface showing assigned tasks and task status.

71. Employee Messages Interface

The My Messages interface allows employees to view internal messages sent by the Manager or administration. The message data is stored in the employee_messages table.

Each message includes a message ID, sender, message content, sent date, and status. The status shows whether the message is Read or Unread. This helps the employee know if there are new messages that need attention.

The interface includes buttons such as Refresh, MarkRead, Reply, and Logout. The Refresh button reloads messages from the database. The MarkRead button changes the selected message status to Read. The Reply button allows the employee to send a response, which is also stored in the database.

This interface improves communication inside the company and makes it easier for employees to receive updates from management.

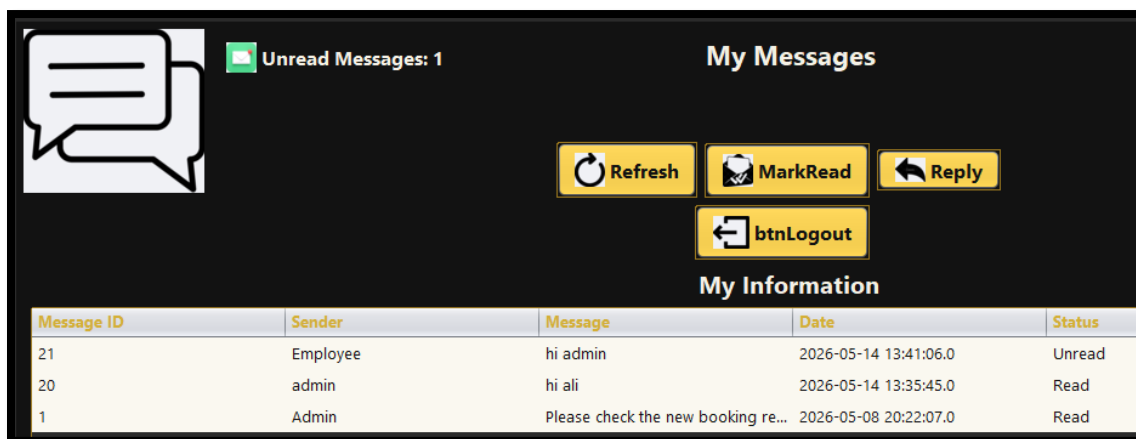


Figure: Employee messages interface for internal communication.

72. Employee Task Status Tracking

Task status tracking is an important part of the Employee Dashboard. Each task has a status value that shows whether it is still pending or completed. The system uses these status values to count pending and completed tasks.

When the employee marks a task as done, the database is updated immediately. After that, the task table and task counters are refreshed. This allows the employee to see the updated task status directly in the interface.

This feature helps employees organize their work and helps the company track task completion. It also shows how the graphical interface and the database work together.

Before :

	task_id	emp_id	title	description	status	due_date
▶	1	1	Check bookings	Review customer bookings for today	Done	2026-05-11
	2	1	Review hotel bookings	Check hotel reservations and room types	Done	2026-05-21
	3	1	a	b	Pending	2020-05-25
*	NULL	NULL	NULL	NULL	NULL	NULL

After :

	task_id	emp_id	title	description	status	due_date
▶	1	1	Check bookings	Review customer bookings for today	Done	2026-05-11
	2	1	Review hotel bookings	Check hotel reservations and room types	Done	2026-05-21
	3	1	a	b	Done	2020-05-25
*	NULL	NULL	NULL	NULL	NULL	NULL

73. Employee Internal Communication

The internal messaging feature allows communication between management and employees inside the system. Instead of using an external communication method, the Manager can send a message directly from the Employees Data Management interface, and the employee can read it from the Messages interface.

Each message is linked to an employee using emp_id. This ensures that each employee sees only their own messages. The system also supports message status values such as Read and Unread, which helps employees track new messages.

This feature makes the system more practical because it does not only store data but also supports internal communication between users.

Hajj Report

I implemented internal messaging between admin and employees

Send Message

Send



Unread Messages: 1

My Messages

Refresh

MarkRead

Reply

btnLogout

My Information

Message ID	Sender	Message	Date	Status
21	Employee	hi admin	2026-05-14 13:41:06.0	Unread
20	admin	hi ali	2026-05-14 13:35:45.0	Read
1	Admin	Please check the new booking re...	2026-05-08 20:22:07.0	Read

Figure: Internal messaging between Manager and Employee.

74. Employee Dashboard Summary

The Employee Dashboard provides work-related services for employees. It includes profile management, task tracking, and internal messages. These features allow employees to stay updated with their work and communicate with management.

The dashboard is connected mainly to the employees, employee_tasks, and employee_messages tables. It uses SQL queries to retrieve employee data, update task status, and display messages.

Overall, the Employee Dashboard supports the daily work of employees and keeps their information organized.

75. Customer Dashboard Overview

The Customer Dashboard is designed for customers or Hajj applicants who use the system to view their Hajj-related services. The dashboard provides access to personal profile information, booking details, transport information, and hotel reservation information.

The Customer Dashboard contains a side menu with buttons such as My Profile, My Booking, My Transport, My Hotel, and Logout. This menu makes it easy for the customer to move between different services.

The customer has limited access compared to the Manager and Employee. The customer cannot manage company records, employees, campaigns, or service providers. Instead, the customer can view only the information related to their own Hajj application and services.

Hajj Report

This dashboard supports the role-based access design of the system and helps customers follow their services in an organized way.

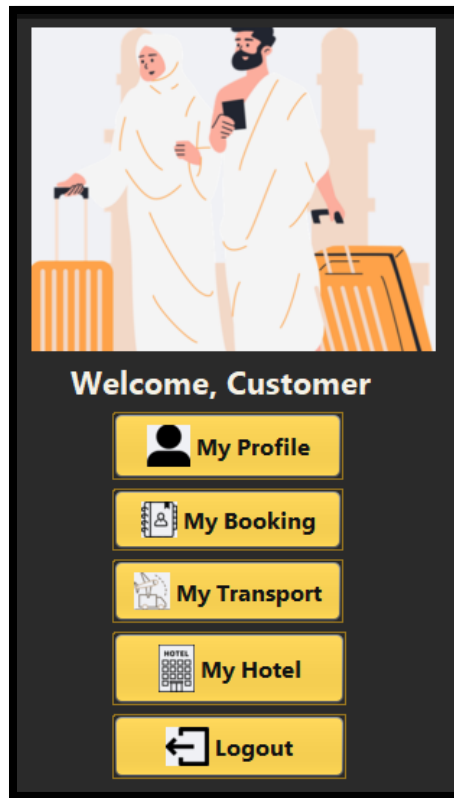


Figure: Customer Dashboard navigation menu.

76. Customer My Profile Interface

The Customer My Profile interface displays the personal Hajj information of the customer. This information is retrieved mainly from the hajj_applicants table. It includes the Hajj applicant ID, first name, last name, gender, date of birth, phone number, and application status.

The interface also displays supervisor information and country information. The supervisor information can include supervisor name, email, phone number, and experience years. The country information can include country name, currency, language, and continent.

The customer can only edit limited information, such as phone number and password. This is important because sensitive information should not be changed freely by the customer. Other information is managed by the system or by the Manager.

This interface gives the customer a clear view of their registered Hajj information and related details.

My Profile

 [Edit Profile](#) [Change Password](#)

[Logout](#)

You can only edit Phone and Password

Hajji Information	Supervisor Information	Country Information
1 mohammed jaber Male 2000-01-01 45454545 Pending	osama supervisor@example.com 599000000 5	Palestine ILS/JOD Arabic Asia

Figure: Customer profile interface displaying Hajj applicant information.

77. Customer Booking Interface

The Customer Booking interface allows the customer to view booking information related to a Hajj package. The interface displays booking ID, package name, start date, end date, booking status, and total price.

This interface is mainly connected to the bookings and packages tables. The packages table stores package details such as package name, start date, end date, and price. The bookings table stores the customer reservation and connects the Hajj applicant with a specific package.

The difference between a package and a booking is important. A package is an available Hajj offer created by the company. A booking is a customer reservation for one of these packages. Therefore, adding a new package alone does not make it appear in the customer booking interface unless there is a booking connected to that package.

The Refresh button reloads the latest booking data from the database. The Cancel Booking button updates the booking status to Cancelled instead of deleting the record completely. This keeps the booking history stored in the database.

My Booking

 Refresh [Cancel Booking](#)

[Logout](#)

My Information

Booking ID	Package Name	Start Date	End Date	Status	Total Price
1	Standard Hajj Package	2026-06-01	2026-06-15	cancelled	2500.00

Figure: Customer booking interface showing package reservation details.

78. Customer Transport Interface

The Customer Transport interface displays transportation information assigned to the customer. The interface shows transport ID, company name, vehicle type, starting location, destination, departure date, and transport status.

This interface is connected to the hajji_transports, transports, and transport_companies tables. The transport_companies table stores company information. The transports table stores trip details. The hajji_transports table connects the Hajj applicant with a specific transport trip.

This structure is important because adding a transport trip to the transports table does not automatically assign it to a customer. The trip must be linked to the customer through the hajji_transports table.

The Refresh button reloads the assigned transport information from the database. The Cancel Transport button can update the status to Cancelled while keeping the record stored for history and tracking.

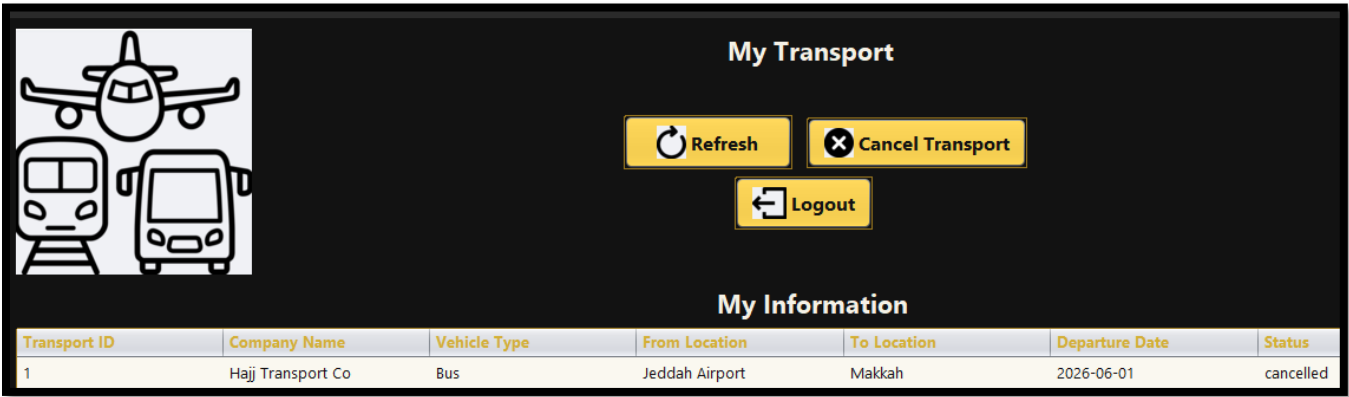


Figure: Customer transport interface showing assigned transport details.

79. Customer Hotel Interface

The Customer Hotel interface displays hotel reservation information for the customer. It shows hotel ID, hotel name, city, room type, check-in date, check-out date, reservation status, and price.

This interface is connected mainly to the hotel_bookings, hotels, and hajj_applicants tables. The hotels table stores general hotel information such as hotel name, phone number, city, address, and rating. The hotel_bookings table stores customer-specific reservation information such as room type, check-in date, check-out date, status, and price.

This design separates hotel details from reservation details. The hotel information is stored once in the hotels table, while each customer reservation is stored in hotel_bookings.

Hajj Report

The Refresh button reloads the latest hotel reservation data. The Cancel Hotel button can update the reservation status to Cancelled instead of removing the record completely.

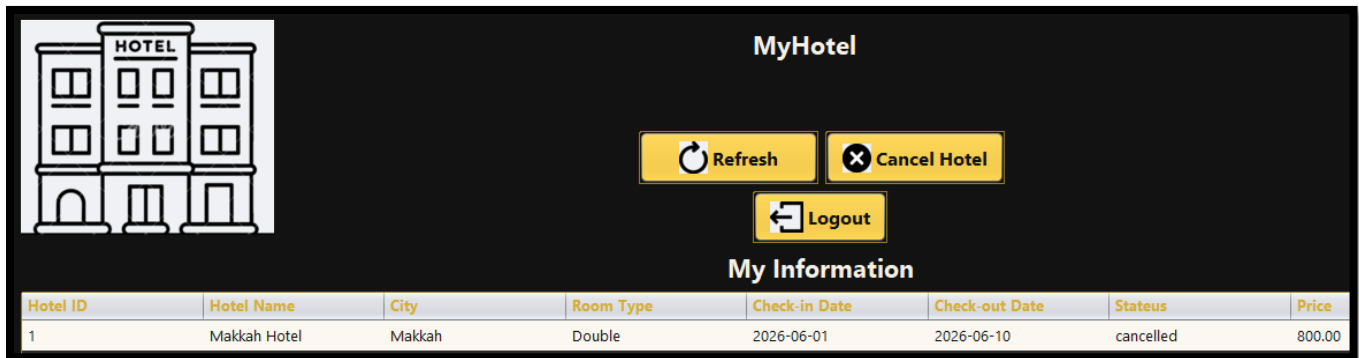


Figure: Customer hotel interface showing hotel reservation details.

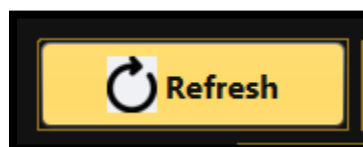
80. Refresh Buttons in Customer Interfaces

The Customer Dashboard contains Refresh buttons in several interfaces, such as My Booking, My Transport, and My Hotel. The purpose of the Refresh button is to reload the latest data from the database.

For example, if a booking status is updated in the database, the customer can click Refresh to see the latest status in the interface. The same idea applies to transport and hotel information.

However, the Refresh button only reloads data that is already connected to the logged-in customer. For example, adding a new package to the packages table will not make it appear in the customer booking screen unless a booking record connects that package to the customer.

This behavior is correct because the customer interfaces show customer-specific data, not all available records in the system.



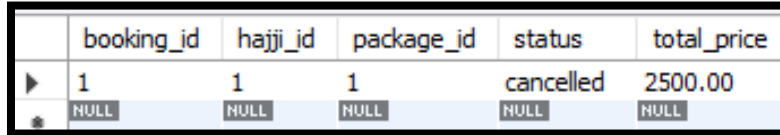
81. Cancel Actions in Customer Interfaces

The Customer Dashboard includes cancel actions such as Cancel Booking, Cancel Transport, and Cancel Hotel. These actions are used when the customer wants to cancel a service.

In many cases, the system does not delete the record completely. Instead, it updates the status to Cancelled. This is better because it keeps the service history in the database. The company can still know that the customer had a booking or service that was later cancelled.

Hajj Report

For example, when a customer cancels a booking, the booking record remains in the bookings table, but its status changes to Cancelled. This approach is useful for reporting and tracking past customer actions.



	booking_id	hajji_id	package_id	status	total_price
▶	1	1	1	cancelled	2500.00
✱	NULL	NULL	NULL	NULL	NULL

Figure: Cancel action updates status instead of deleting the record.

82. Customer Dashboard Summary

The Customer Dashboard provides customer-related services in an organized way. It allows the customer to view profile information, booking details, transport details, and hotel reservation information.

The dashboard is connected to several tables, including hajj_applicants, bookings, packages, hotel_bookings, hotels, hajji_transports, transports, and transport_companies.

This dashboard supports limited customer access and ensures that customers only see their own information. It also helps the company provide a clear and organized service experience for customers.

83. CRUD Operations in the System

CRUD operations are the basic database operations used in the system. CRUD stands for Create, Read, Update, and Delete. These operations allow the system to add new data, display existing data, modify records, and remove records when needed.

In this project, CRUD operations are mainly used in the Manager Dashboard. The Manager can add new campaigns, employees, hotels, transport companies, and Hajj applicants. The Manager can also update existing records, delete records, and view data inside tables.

These operations are connected directly to SQL commands. Create is done using INSERT, Read is done using SELECT, Update is done using UPDATE, and Delete is done using DELETE.



Hajj Report

84. Create Operation

The Create operation is used to add new records to the database. In the system, this operation happens when the Manager clicks the Add button in management screens.

For example, when a new employee is added, the system takes the values from the text fields and sends an INSERT query to the employees table. After the data is inserted, the table is refreshed to show the new record.

This operation is important because it allows the company to add new information to the system as needed.

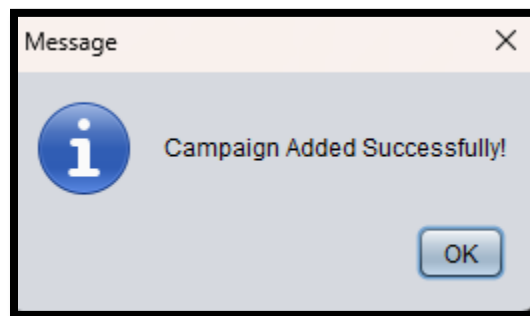
Campaigns Management

AddUpdateDeleteClear

ID: Name: Location:
Email:

Data description table

ID	Name	location	Email
1	Main Hajj Campaign	Makkah	campaign@example.com



Campaigns Management

AddUpdateDeleteClear

ID: Name: Location:
Email:

Data description table

ID	Name	location	Email
1	Main Hajj Campaign	Makkah	campaign@example.com
12	a	a	a@gmail

Hajj Report

85. Read Operation

The Read operation is used to retrieve and display data from the database. Most interfaces in the system use this operation to show records inside JTable.

For example, the Manager can view all employees, hotels, campaigns, and Hajj applicants. The Employee can view tasks and messages. The Customer can view booking, hotel, and transport information.

This operation is implemented using the SQL SELECT command.

	task_id	emp_id	title	description	status	due_date
▶	1	1	Check bookings	Review customer bookings for today	Done	2026-05-11
	2	1	Review hotel bookings	Check hotel reservations and room types	Done	2026-05-21
	3	1	a	b	Done	2020-05-25
★	NULL	NULL	NULL	NULL	NULL	NULL

86. Update Operation

The Update operation is used to modify existing records. In the Manager Dashboard, the Manager can select a row from a table, edit the fields, and click Update to save the changes.

The Employee also uses update operations when marking a task as Done. The Customer uses update operations when editing the phone number or changing the password.

This operation is implemented using the SQL UPDATE command.

Campaigns Management

AddUpdateDeleteClear

ID: Name: Location:
Email:

Data description table

ID	Name	location	Email
1	Main Hajj Campaign	Makkah	campaign@example.com
12	a	a	a@gmail

Message

Campaign Updated!

OK

Campaigns Management

AddUpdateDeleteClear

ID: Name: Location:
Email:

ID	Name	location	Email
1	ZX		
12	a	a	a@gmail

87. Delete Operation

The Delete operation is used to remove records from the database. In the Manager Dashboard, delete operations are available in management screens such as Campaigns, Employees, Hotels, and Transport Companies.

However, some customer actions do not delete records completely. For example, Cancel Booking may change the status to Cancelled instead of deleting the booking. This keeps the history of the service in the database.

This operation is implemented using the SQL DELETE command, while cancellation is usually implemented using UPDATE.

Campaigns Management

AddUpdateDeleteClear

ID: Name: Location:
Email:

ID	Name	location	Email
1	ZX		
12	a	a	a@gmail

Message

 Campaign Deleted!

OK

Campaigns Management

AddUpdateDeleteClear

ID: Name: Location:
Email:

Data description table

ID	Name	location	Email
12	a	a	a@gmail

88. Testing and Validation

The system was tested using sample data stored in the MySQL database. Testing was done by using the graphical interfaces and checking that the database was updated correctly. For example, adding a new record from the Manager Dashboard should appear in the related database table. Updating a task status should change the value in the employee_tasks table. Cancelling a booking should update the booking status to Cancelled.

The testing process confirmed that the Java interfaces are correctly connected to the MySQL database and that the main operations work as expected.

	task_id	emp_id	title	description	status	due_date
▶	1	1	Check bookings	Review customer bookings for today	Done	2026-05-11
	2	1	Review hotel bookings	Check hotel reservations and room types	Done	2026-05-21
	3	1	a	b	Pending	2020-05-25
✱	NULL	NULL	NULL	NULL	NULL	NULL

89. Challenges and Solutions

During the development of the system, some challenges appeared. One challenge was connecting the Java application with the MySQL database correctly. This was solved by using JDBC and a separate DBConnection.java class. Another challenge was displaying the correct data for each logged-in user. This was solved by using IDs such as emp_id and hajji_id to filter the data.

The project also required organizing many tables and relationships. This was solved by using primary keys, foreign keys, and normalized table design.

90. Future Improvements

The system can be improved in the future by adding more advanced features. These features may include search and filtering options, report generation, stronger password security, notifications, and better dashboard design. The system can also be improved by adding more validation for user input and adding more detailed reports for bookings, hotels, and transport services.

91. Conclusion

In conclusion, the Al-Usra Hajj and Umrah Company Management System provides an organized database solution for managing the main operations of a Hajj and Umrah company. The system connects Java Swing interfaces with a MySQL relational database and supports different user roles: Manager, Employee, and Customer.

The project applies important database concepts such as tables, relationships, primary keys, foreign keys, constraints, normalization, CRUD operations, and JDBC connectivity. Overall, the system helps reduce manual work, organize company data, and make Hajj and Umrah service management easier and more efficient.

Project Resources

The project files can be accessed using the following links:

Source Code:

<https://github.com/HibaMeneh/HajiiCompanySystem>

Database SQL File:

<https://drive.google.com/drive/folders/1SfC1LHnourozUQZdVeOvd1eFaKZ-j73z?hl=ar>

Project Report:

<https://drive.google.com/drive/folders/18duxPOWe-xY3yPafysyipHD4weGxBhdY?hl=ar>

ERD and Normalization Document:

https://drive.google.com/drive/folders/1ctKbE_jBrty-wuRPUunhvnDcrDKsRIEY?hl=ar